```
In[ ]:= SetOptions[$FrontEndSession, DynamicEvaluationTimeout → 3000];
        SetOptions[$FrontEndSession, DynamicUpdating → False];
        SetOptions[$FrontEnd, DynamicUpdating → False];
        SetOptions[EvaluationNotebook[], DynamicUpdating → False];

In[ ]:= CurrentValue[DynamicUpdating]       (*Should give "False"*)

Out[ ]=
        False
```

Solutions to second post-Newtonian dynamics of spinning and eccentric binary black holes (accompanying arXiv:2603.20031)

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A. README: Post-Newtonian Dynamics Notebook User Manual

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1. Physics Definitions

Before configuring the code, note the underlying Hamiltonian terms:
$h_{\text{full}} = h_N + h_{1\,\text{PN}} + h_{\text{so}} + (h_2 + h_{\text{ss}})$

- h_N = Pure Newtonian gravity
- h_1, h_2 = 1PN and 2PN purely orbital corrections
- h_{so} = 1.5PN Spin-Orbit coupling
- h_{ss} = 2PN Spin-Spin coupling

2. Physics Models & Plotting Indices

We can plot the phase space variable solution under various evolution schemes. Use these 7 indices in the graphing command DefPlot below in Sec. B-6-B to select the evolution scheme.
Note: All models compute non-orbit averaged dynamics, except Index 3 which computes 'Orbit-Averaged' dynamics.

Index	Physics Model Plotted	Hamiltonian Equation	Code Routine
1	1.5 PN Spin – Orbit (Analytic)	$H = h_N + h_1 + h_{so}$	Hflow1List
2	2 PN Spin – Spin (Analytic)	$H = h_N + h_1 + h_{so} + h_2 + h_{ss}$	Hflow2SList
3	2 PN Spin – Spin Orbit – averaged (Analytic)	$H = h_N + h_1 + h_{so} + h_2 + h_{ss}$	Hflow2SList
4	1.5 PN Spin – Orbit (Numerical)	$H = h_N + h_1 + h_{so}$	ComputeNum1
5	2 PN Spin – Spin (Numerical)	$H = h_N + h_1 + h_{so} + h_2 + h_{ss}$	ComputeNum2S
6	2 PN Pure Orbital (Analytic)	$H = h_N + h_1 + h_2$ (Spins = 0)	Hflow2SListFixedSpins
7	2 PN Pure Orbital (Numerical)	$H = h_N + h_1 + h_2$ (Spins = 0)	ComputeNumOrb

3. Initialization

Go to Sec. B-1 below and set things up.

- Orb (for analytical solutions) and NumOrb (for numerical solutions): Toggles the purely orbital corrections (h_1 and h_2). Default is "Yes". If set to "No", the purely orbital parts of the Hamiltonian (without spins) h_1 and h_2 are ignored, and equations of motion change accordingly.
- QKP: This is valid only for “Index 1” solution mentioned in the above table. It decides which baseline solution for $|\mathbf{R}| = |\boldsymbol{R_1} - \boldsymbol{R_2}|$ (orbital separation vector norm) is to be injected for computing spin solutions; see Eq. 3.5 of https://arxiv.org/abs/1908.02927. In this paper, the Newtonian solution was injected. We have two options. Choosing “N” injects the Newtonian solution for $|\mathbf{R}|$ (just like https://arxiv.org/abs/1908.02927), whereas choosing “SO” injects the 1.5PN solution for $|\mathbf{R}|$. Our default value is “SO”.

- Torb: The rough orbital time period of the system, the approximate time in which $|\mathbf{R}|$ returns back to its original value.
- end[data]: Total numerical integration runtime and plot display time.
- userData: Your initial data formatted as $\{m_1, m_2, \mathbf{R}, \mathbf{P}, \boldsymbol{S_1}/\epsilon^{1/2}, \boldsymbol{S_2}/\epsilon^{1/2}, \epsilon = 1/c^2\}$. Here $\mathbf{P} = \boldsymbol{P_1} = -\boldsymbol{P_2}$, as we are working in the centre-of-mass frame
  - Constraint: m_1 must be strictly greater than m_2.
  - Constants: G = 1.

4. Setting up Plotting Parameters

Then go to Sec. B-6-B to set the plotting parameters, which we now explain. Here we plot using the function DefPlot

```
Syntax: DefPlot[data, vector_number, component_number, time_min,
time_max, {plot_indices}]
```

- vector_number: 1 ($\mathbf{R}$), 2 ($\mathbf{L}$), 3 ($\boldsymbol{S_1}$), or 4 ($\boldsymbol{S_2}$).
- component_number: 1 (x-component), 2 (y-component), or 3 (z-component).

Example: DefPlot[data, 2, 3, tmin, tmax, {2, 3, 5}] graphs the z-component of Angular Momentum overlaying Analytical, Averaged, and Numerical 2PN.

5. Succinct Overview: How to Run the Notebook

- Set parameters up in Secs. B-1 and B-6-B (to be found below), as explained above.
- Quit the kernel and clear all memory (Evaluation --> Quit Kernel).
- Run the first cell (at the very top of the notebook) *manually* (i.e. via Shift + Enter), which has the following content: “SetOptions[$FrontEndSession ...”
- Run the entire notebook (Evaluation --> Evaluate Notebook). That’s it!
- Sec. B-4 generates a **summary table** for the system with its important parameters displayed: **(a)** mass ratio q **(b)** angles b/w angular momenta $\kappa 1$ (L and S_1), $\kappa 2$ (L and S_1), γ (S_1 and S_2) **(c)** PN parameter PNa = $G\,M/(c^2\,R_{\text{initial}})$, which does not take eccentricity into account **(d)** PN parameter PNp = $G\,M/(c^2\,a_N\,(1-e_N))$, which takes eccentricity into account. Here M = total mass, G =1, e_N = Newtonian eccentricity, a_n = Newtonian semi major axis for the unscaled R. In this table, are also displayed **(e)** dimensionless spins χ_1 and χ_2 **(f)** $S_{1\,\text{or}\,2}/(\epsilon^{1/2}\,L)$, plus other QKP parameters like scaled Hamiltonian, radial eccentricity, mean motion, orbital time period and spin-precession time period.

- **IMPORTANT: Our solutions work well when (a)** both PN parameters are below sub-percent level, **(b)** eccentricity is less than and away from 1, **(c)** $\chi_{1\,\mathrm{or}\,2}$ is less than and away from 1, **(d)** $S_{1\,\mathrm{or}\,2}/(L\,\epsilon^{1/2})$ is less than and away from 1. **(e)** Scaled Hamiltonian h < 0, **(f)** κ_1, κ_2, γ are not 0 or π radians (away from spin aligned or anti-aligned cases). The values of all these parameters are found in the so-called “summary table” discussed above.

B. Code begins

1. Set Parameters

In[∘]:=
```
Orb = "Yes";
NumOrb = "Yes";
QKP = "SO";
end[data_] :=  600 Torb[data]

(*Input initial data in the
 format: {m1,m2,R,P,S1/ϵ^(1/2),S2/ϵ^(1/2),ϵ = 1/c^2}.  Keep m1 > m2 *)
userData = {3.5, 2.3, {-3, -5, 3},
   -1 / 2 {2, 1, 2}, -1 {-2, -3, -2} , -1 / 2 {-4, -3, -1} , 0.008};

accuracyGoal = 14;
precisionGoal = 14;
maxSteps = Infinity;
workingprecision = 40;
```

2. Analytical Solution

A. HN + H1PN + HSO (QKP = “N” or “SO”) & (ORB = “Yes” or “No”)

In[∘]:=
```
(* ====================*)(*GEOMETRY& EULER MATRICES*)
(* ==========================*)
sphericalAngles[V_] :=
  Module[{polarAngJ, azimuthAngJ}, polarAngJ = ToSphericalCoordinates[Re[V]][[2]];
   azimuthAngJ = ToSphericalCoordinates[Re[V]][[3]];
   Return[{polarAngJ, azimuthAngJ}];];

vectorComponents[V_, θ_, φ_] := V * {Sin[θ] * Cos[φ], Sin[θ] * Sin[φ], Cos[θ]};
```

```
computeEulerMatrix[jvec_] :=
  Module[{θ, ϕ, cosθ, sinθ, cosϕ, sinϕ}, {θ, ϕ} = sphericalAngles[jvec] + {0, π / 2};
   cosθ = Cos[θ]; sinθ = Sin[θ];
   cosϕ = Cos[ϕ]; sinϕ = Sin[ϕ];
   N[{{cosϕ, sinϕ, 0},
     {-sinϕ * cosθ, cosϕ * cosθ, sinθ}, {sinϕ * sinθ, -cosϕ * sinθ, cosθ}}]];

(* ==============================*)(*BASE QUANTITIES COMPUTATION (t=0)*)
(* ========================================*)
computeBaseQuantities1[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, QKP_, ORB_] :=
  Module[{o, c, M, μ, ν, rvec, pvec, r, p, SS10, s1vec, s1, SS20, s2vec,
    s2, lvec, l, jvec, j, EulMat, InvEulMat, δ1, δ2, seffdl, h, h0,
    Σ1, Σ2, A3, A2, A1, A0, x2, x3, x1, β, SqA, JtoLframeEulMat0,
    n, nunscaled, nso, nunscaledso, ar,
    arso, er, erso, et, etso, eϕ, eϕso, fP, gP, fϕ, gϕ, Φ, d,
    α1l, α2l, β1l, β2l, α1s1, α2s1, β1s1, β2s1, βeϕso, βerso, βeϕ, βer,
    drdt0, u0testso, Absu0so,
    u0so, ver0so, veϕ0so, t0so, sign, cosκ10, Int30, α, ϒ0,
    lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0, s1ξ10,
    solPieceS110, solPieceS120, r0Lframe, rϕ0, solPieceR10, solPieceR20,
    Nprec, Uprec, Tprec, Torb, freqξ1L, freqcosκ1}, c = 1 / Sqrt[ϵ];

   If[ORB == "Yes", o = 1]; If[ORB == "No", o = 0];

   M = N[m1 + m2]; μ = N[m1 m2 / M]; ν = μ / M;
   rvec = N[R0 / M]; r = Norm[rvec];
   pvec = N[P0 / μ]; p = Norm[pvec];
   SS10 = N[S10 Sqrt[ϵ]]; s1vec = SS10 / (μ M); s1 = Norm[s1vec];
   SS20 = N[S20 Sqrt[ϵ]]; s2vec = SS20 / (μ M); s2 = Norm[s2vec];
   lvec = Cross[rvec, pvec]; l = Norm[lvec];
   jvec = lvec + s1vec + s2vec; j = Norm[jvec];
   δ1 = 2 ν (1 + 3 m2 / (4 m1));
   δ2 = 2 ν (1 + 3 m1 / (4 m2));
   seffdl = Dot[lvec, δ1 s1vec + δ2 s2vec];

   h = p^2 / 2 - 1 / r + o ϵ (1 / (2 r^2) + 1 / 8 p^4 (-1 + 3 ν) -
         (Dot[rvec / r, pvec]^2 ν + p^2 (3 + ν)) / (2 r)) + (seffdl ϵ) / r^3;
   h0 = p^2 / 2 - 1 / r;
   EulMat = computeEulerMatrix[jvec]; InvEulMat = Inverse[EulMat];

   Σ1 = N[s1vec.s2vec / (s1 s2) - l / s2 (δ1 - δ2) / δ2 s1vec.lvec / (l s1)];
   Σ2 = N[s2vec.lvec / (l s2) + (δ1 s1) / (δ2 s2) s1vec.lvec / (l s1)];

   A3 = 2 l s1 δ1 (δ2 - δ1);
```

```
A2 =
 -(l^2 (δ1 - δ2)^2 + 2 δ2 l Σ2 s2 (δ2 - δ1) + s1^2 δ1^2 + 2 δ1 δ2 Σ1 s1 s2 + δ2^2 s2^2);
A1 = (2 δ2 s2 (l Σ1 (δ2 - δ1) + Σ2 (δ1 s1 + δ2 Σ1 s2)));
A0 = -s2^2 δ2^2 (Σ1^2 + Σ2^2 - 1);
{x2, x3, x1} = Sort[x /. Solve[A3 x^3 + A2 x^2 + A1 x + A0 == 0, x]];
β = Sqrt[(x3 - x2) / (x1 - x2)];
SqA = 2 / Sqrt[A3 (x1 - x2)];
JtoLframeEulMat0 = computeEulerMatrix[EulMat.lvec];

(*1. Unconditionally compute
 strictly SO (1.5PN) variables for main orbit geometry*)
nso = 2 Sqrt[2] (-h)^(3/2) - o ((-h)^(3/2) h ϵ (-15 + ν)) / Sqrt[2];
nunscaledso = nso / M;
arso = -(1 / (2 h)) + o 1/4 ϵ ((-7 + ν) + (4 seffdl) / l^2);
erso = Sqrt[
  1 + 2 h l^2 + o h ϵ (2 (-6 + ν) + 5 h l^2 (-3 + ν)) + (8 h (1 + h l^2) seffdl ϵ) / l^2];
etso = Sqrt[1 + 2 h l^2 + o ϵ (4 h (1 - ν) + h^2 l^2 (17 - 7 ν)) + (4 h seffdl ϵ) / l^2];
eϕso = Sqrt[1 + 2 h l^2 + o ϵ (-12 h + h^2 l^2 (-15 + ν)) +
   ϵ 4 (δ1 + δ2) (1 + 2 h l^2) h + 4 ϵ (3 + 4 h l^2) seffdl h / l^2];

(*2. Conditionally compute the
 variables strictly for use INSIDE Upsilon (Int30)*)
n = If[QKP == "N", 2 Sqrt[2] (-h0)^(3/2), nso];
nunscaled = n / M;
ar = If[QKP == "N", -(1 / (2 h0)), arso];
er = If[QKP == "N", Sqrt[1 + 2 h0 l^2], erso];
et = If[QKP == "N", Sqrt[1 + 2 h0 l^2], etso];
eϕ = If[QKP == "N", Sqrt[1 + 2 h0 l^2], eϕso];
fP = 0; gP = 0; fϕ = 0; gϕ = 0;
Φ = 1 + o ϵ 3 / l^2 - ϵ / l^4 (l^2 δ1 + l^2 δ2 + 3 seffdl);

d = ((2 + er^2 - 3 er et) / (2 ar^3 (1 - er^2)^(5/2)))^(-1/3);
α1l = -((δ2 (j + l + s2 Σ2)) / (s1 (δ1 - δ2)));
α2l = -((δ2 (-j + l + s2 Σ2)) / (s1 (δ1 - δ2)));
β1l = -(δ2 / (2 s1 (δ1 - δ2)))
  (l^2 δ1 + j^2 δ2 + s1 (δ1 - δ2) (s1 + s2 Σ1) + l s2 δ1 Σ2 + j (l (δ1 + δ2) + s2 δ2 Σ2));
β2l = -(δ2 / (2 s1 (δ1 - δ2)))
  (l^2 δ1 + j^2 δ2 + s1 (δ1 - δ2) (s1 + s2 Σ1) + l s2 δ1 Σ2 - j (l (δ1 + δ2) + s2 δ2 Σ2));
α1s1 = δ2 (-j + s1 + s2 Σ1) / (l δ1);
α2s1 = δ2 (j + s1 + s2 Σ1) / (l δ1);
β1s1 = -1 / (2 l δ1) δ2 (l^2 δ1 + (s1 (δ1 - δ2) + j δ2) (-j + s1 + s2 Σ1) + l s2 δ1 Σ2);
β2s1 = 1 / (2 l δ1) δ2 (-l^2 δ1 + (-s1 δ1 + (j + s1) δ2) (j + s1 + s2 Σ1) - l s2 δ1 Σ2);
drdt0 =
 Sign[1 / (8 r^4) pvec.rvec (-o 4 r^2 ϵ (6 + 4 ν) + r^3 (8 + o 4 p^2 ϵ (-1 + 3 ν)))];
```

```
(* ================*)(*3. Initial parameters*)(* ==================*)
βeϕso = eϕso / (1 + Sqrt[1 - eϕso ^ 2]);
βerso = erso / (1 + Sqrt[1 - erso ^ 2]);
u0testso = ArcCos[Clip[(r / arso - 1) / (-erso), {-1, 1}]];
Absu0so = u /. FindRoot[r == arso (1 - erso Cos[u]), {u, u0testso}];
u0so = drdt0 * Absu0so;
ver0so = u0so + 2 ArcTan[(βerso Sin[u0so]) / (1 - βerso Cos[u0so])];
veϕ0so = u0so + 2 ArcTan[(βeϕso Sin[u0so]) / (1 - βeϕso Cos[u0so])];
t0so = - (u0so - etso Sin[u0so]) / nunscaledso;

βeϕ = eϕ / (1 + Sqrt[1 - eϕ ^ 2]);
βer = er / (1 + Sqrt[1 - er ^ 2]);
sign = If[Dot[lvec, Cross[s1vec, s2vec]] > 0, 1, -1];
cosκ10 = Dot[lvec, s1vec] / (l s1);

Int30 = (1 / (2 ar ^ 3 n)) * (((2 + er ^ 2 - 3 er et) ver0so) / (1 - er ^ 2) ^ (5 / 2) +
    ((-er + et) Sin[u0so]) / ((-1 + er ^ 2) (-1 + er Cos[u0so]) ^ 2) +
    ((-3 er + et + 2 er ^ 2 et) Sin[u0so]) / ((-1 + er ^ 2) ^ 2 (-1 + er Cos[u0so])));

α = sign * SqA * EllipticF[ArcSin[Sqrt[(cosκ10 - x2) / (x3 - x2)]], β ^ 2] - (ϵ * Int30);
ϒ0 = 1 / SqA * (α + ϵ * Int30);
lvecJframe0 = EulMat.lvec;
lξ10 = sphericalAngles[lvecJframe0][[2]] + π / 2;
solPieceL10 = SqA (β1l
     EllipticPi[(x2 - x3) / (α1l + x2), JacobiAmplitude[ϒ0, β ^ 2], β ^ 2] / (α1l + x2)
    - β2l EllipticPi[(x2 - x3) / (α2l + x2),
       JacobiAmplitude[ϒ0, β ^ 2], β ^ 2] / (α2l + x2));
solPieceL20 = lξ10 - solPieceL10;

s1vecJframe0 = EulMat.s1vec;
s1ξ10 = sphericalAngles[s1vecJframe0][[2]] + π / 2;
solPieceS110 = SqA (β1s1 EllipticPi[
        (x2 - x3) / (α1s1 + x2), JacobiAmplitude[ϒ0, β ^ 2], β ^ 2] / (α1s1 + x2)
    - β2s1 EllipticPi[(x2 - x3) / (α2s1 + x2),
       JacobiAmplitude[ϒ0, β ^ 2], β ^ 2] / (α2s1 + x2)
    + j δ2 ϒ0);
solPieceS120 = s1ξ10 - solPieceS110;

r0Lframe = JtoLframeEulMat0.EulMat.rvec;
rϕ0 = sphericalAngles[r0Lframe][[2]];
solPieceR10 = SqA (β1l
     EllipticPi[(x2 - x3) / (α1l + x2), JacobiAmplitude[ϒ0, β ^ 2], β ^ 2] / (α1l + x2)
    + β2l EllipticPi[(x2 - x3) / (α2l + x2),
```

```
            JacobiAmplitude[ϒ0, β^2], β^2] / (α2l + x2));
    solPieceR20 = Φ veϕ0so;

    Nprec = Hypergeometric2F1[1 / 2, 1 / 2, 1, β^2]
      (2 ar^3 n (1 - er^2)^(5 / 2)) / (ϵ Sqrt[A3 (x1 - x2)] (2 + er^2 - 3 et er));
    Uprec = 2 π Nprec;
    Tprec = Uprec / n M;
    Torb = 2 π / nunscaled;
    freqξ1L =
     (ϵ ((β1l EllipticPi[(x2 - x3) / (x2 + α1l), β^2]) / (x2 + α1l) - (β2l EllipticPi[
              (x2 - x3) / (x2 + α2l), β^2]) / (x2 + α2l))) / (d^3 M EllipticK[β^2]);
    freqcosκ1 = nunscaled / Nprec;

    Return[
     N[{c, M, μ, ν, rvec, pvec, r, p, SS10, s1vec, s1, SS20, s2vec, s2, lvec, l, jvec,
       j, EulMat, InvEulMat, δ1, δ2, seffdl, h, h0, Σ1, Σ2, A3, A2, A1, A0, x2, x3, x1,
       β, SqA, JtoLframeEulMat0, n, nunscaled, nso, nunscaledso, ar, arso, er, erso,
       et, etso, eϕ, eϕso, fP, gP, fϕ, gϕ, Φ, d, α1l, α2l, β1l, β2l, α1s1, α2s1, β1s1,
       β2s1, βeϕso, βerso, βeϕ, βer, drdt0, t0so, sign, cosκ10, Int30, α,
       ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0,
       s1ξ10, solPieceS110, solPieceS120, r0Lframe, rϕ0, solPieceR10,
       solPieceR20, Nprec, Uprec, Tprec, Torb, freqξ1L, freqcosκ1}]];];


(* ==================================================================*)
(*TIME PROPAGATION MODULE*)
(* ==================================================================*)
Hflow1List[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, QKP_, ORB_, tList_] :=
  Module[{pi, c, M, μ, ν, rvec, pvec, r, p, SS10, s1vec, s1, SS20, s2vec,
    s2, lvec, l, jvec, j, EulMat, InvEulMat, δ1, δ2, seffdl, h, h0,
    Σ1, Σ2, A3, A2, A1, A0, x2, x3, x1, β, SqA,
    n, nunscaled, nso, nunscaledso, ar,
    arso, er, erso, et, etso, eϕ, eϕso, fP, gP, fϕ, gϕ, Φ, d,
    α1l, α2l, β1l, β2l, α1s1, α2s1, β1s1, β2s1, βeϕso, βerso, βeϕ, βer,
    drdt0, t0so, sign, cosκ10, Int30, α, ϒ0,
    JtoLframeEulMat0, lvecJframe0, lξ10, solPieceL10,
    solPieceL20, s1vecJframe0, s1ξ10, solPieceS110, solPieceS120,
    r0Lframe, rϕ0, solPieceR10, solPieceR20,
    Nprec, Uprec, Tprec, Torb, freqξ1L, freqcosκ1, computeForTime},
   pi = 3.141592653589793`;
   {c, M, μ, ν, rvec, pvec, r, p, SS10, s1vec, s1, SS20, s2vec, s2, lvec, l, jvec,
     j, EulMat, InvEulMat, δ1, δ2, seffdl, h, h0, Σ1, Σ2, A3, A2, A1, A0, x2,
     x3, x1, β, SqA, JtoLframeEulMat0, n, nunscaled, nso, nunscaledso, ar,
     arso, er, erso, et, etso, eϕ, eϕso, fP, gP, fϕ, gϕ, Φ, d, α1l, α2l, β1l,
```

```
  β2l, α1s1, α2s1, β1s1, β2s1, βeϕso, βerso, βeϕ, βer, drdt0, t0so, sign,
  cosκ10, Int30, α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20,
  s1vecJframe0, s1ξ10, solPieceS110, solPieceS120, r0Lframe, rϕ0,
  solPieceR10, solPieceR20, Nprec, Uprec, Tprec, Torb, freqξ1L, freqcosκ1} =
 computeBaseQuantities1[m1, m2, R0, P0, S10, S20, ϵ, QKP, ORB];

computeForTime[tmax_] := Module[{
   finduso, umaxso, cosuso, sinuso,
   veϕso, findu, umax, cosu, sinu, ver, Int3, ϒ, JacAmp, cosκ1,
   solPieceL1, lξ1sol, cosκ2,
   lvecAzimuthAngle, lvecPolarAngle, LsolJframe, Lsol,
   solPieceS11, s1ξ1sol, cosγ,
   s1vecAzimuthAngle, s1vecPolarAngle, S1sol, S2sol,
   JtoLframeEulMat, solPieceR1, solPieceR2, solPieceR3,
   solPieceR4, rvecAzimuthAngleLframe, rMagnitude, Rsol},

  (* ==*)
  (*1. Main Orbit Trajectory:ALWAYS evaluates 1.5PN parameters*)(*==*)
  finduso[t_] := Module[{target, initialGuess}, target = nunscaledso (t - t0so);
    initialGuess = target;
    FindRoot[target == u - etso Sin[u], {u, initialGuess},
     Method → "Newton", WorkingPrecision → MachinePrecision]];
  umaxso = u /. finduso[tmax];
  cosuso = Cos[umaxso]; sinuso = Sin[umaxso];
  veϕso = umaxso + 2. ArcTan[(βeϕso sinuso) / (1 - βeϕso cosuso)];

  (*==*)
  (*2. Isolated Upsilon Trajectory:dynamically tracks QKP choice*)(*==*)
  findu[t_] := Module[{target, initialGuess}, target = nunscaled (t - t0so);
    initialGuess = target;
    FindRoot[target == u - et Sin[u], {u, initialGuess},
     Method → "Newton", WorkingPrecision → MachinePrecision]];
  umax = u /. findu[tmax];
  cosu = Cos[umax]; sinu = Sin[umax];
  ver = umax + 2. ArcTan[(βer sinu) / (1 - βer cosu)];

  (*Int3 is evaluated STRICTLY with the Newtonian QKP if needed*)
  Int3 = (1. / (2. ar^3. n)) * (((2. + er^2. - 3. er et) ver) / (1. - er^2.)^(5. / 2.) +
      ((-er + et) sinu) / ((-1. + er^2.) (-1. + er cosu)^2.) +
      ((-3. er + et + 2. er^2. et) sinu) / ((-1. + er^2.)^2. (-1. + er cosu)));
  ϒ = 1 / SqA (α + ϵ Int3);
  JacAmp = JacobiAmplitude[ϒ, β^2];
  cosκ1 = x2 + (x3 - x2) JacobiSN[ϒ, β^2]^2;
```

```
    (*==*3. SOLUTION FOR VECTOR L==*)
    solPieceL1 = SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), JacAmp, β^2] / (α1l + x2)
        - β2l EllipticPi[(x2 - x3) / (α2l + x2), JacAmp, β^2] / (α2l + x2));
    lξ1sol = solPieceL1 + solPieceL20;
    cosκ2 = Σ2 - δ1 s1 / (δ2 s2) cosκ1;
    lvecAzimuthAngle = -pi / 2. + lξ1sol;
    lvecPolarArgle = ArcCos[(l^2 + l s1 cosκ1 + l s2 cosκ2) / (l j)];
    LsolJframe = μ M vectorComponents[l, lvecPolarAngle, lvecAzimuthAngle];
    Lsol = (InvEulMat.LsolJframe);

    (*==SOLUTION FOR VECTOR S1==*)
    solPieceS11 =
     SqA (β1s1 EllipticPi[(x2 - x3) / (α1s1 + x2), JacAmp, β^2] / (α1s1 + x2)
        - β2s1 EllipticPi[(x2 - x3) / (α2s1 + x2), JacAmp, β^2] / (α2s1 + x2)
        + j δ2 ϒ);
    s1ξ1sol = solPieceS11 + solPieceS120;
    cosγ = Σ1 - l (δ2 - δ1) / (δ2 s2) cosκ1;
    s1vecAzimuthAngle = -pi / 2. + s1ξ1sol;
    s1vecPolarAngle = ArcCos[(s1^2. + l s1 cosκ1 + s1 s2 cosγ) / (s1 j)];
    S1sol =
     μ M (InvEulMat.vectorComponents[s1, s1vecPolarAngle, s1vecAzimuthAngle]);

    (* =====SOLUTION FOR VECTOR S2=====*)
    S2sol = (jvec * μ * M) - Lsol - S1sol;

    (* =====SOLUTION FOR VECTOR R=====*)
    JtoLframeEulMat = computeEulerMatrix[LsolJframe];
    solPieceR1 = SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), JacAmp, β^2] / (α1l + x2)
        + β2l EllipticPi[(x2 - x3) / (α2l + x2), JacAmp, β^2] / (α2l + x2));
    solPieceR2 = Φ veϕso;
    solPieceR3 = solPieceR1 + solPieceR2;
    solPieceR4 = rϕ0 - (solPieceR10 + solPieceR20);
    rvecAzimuthAngleLframe = solPieceR3 + solPieceR4;

    (*Physical Radius physically constrained to 1.5PN*)
    rMagnitude = arso (1 - erso Cos[umaxso]);

    Rsol = M (InvEulMat.Inverse[JtoLframeEulMat].
        vectorComponents[rMagnitude, pi / 2., rvecAzimuthAngleLframe]);
    {(*1*)Rsol, (*2*)Lsol, (*3*)S1sol, (*4*)S2sol, (*5*)Tprec, (*6*)
     Torb, (*7*)rvecAzimuthAngleLframe, (*8*)Transpose[JtoLframeEulMat],
     (*9*)vectorComponents[rMagnitude, pi / 2., rvecAzimuthAngleLframe]}];
Map[computeForTime, tList]];
```

```
(*EXECUTION WRAPPERS*)
DataAn1[QKP_, ORB_, t_, data_] := Block[{rvecAzimuthAngleLframe},
  Module[{m1, m2, R0, P0, S10, S20, ϵ}, {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   Hflow1List[m1, m2, R0, P0, S10, S20, ϵ, QKP, ORB, t]]]

An1[QKP_, ORB_, t_] := Association[# :→ DataAn1[QKP, ORB, t, #] & /@ Keys[Inputs]]
```

B. HN + H1PN + HSO + H2PN + HSS (AVG = “Yes” or “No”) & (ORB = “Yes” or “No”)

```
In[ ]:= (*==*) (*GEOMETRY & EULER MATRICES*) (*==*)
spherical Angles[V_] :=
  Module[{polarAngJ, azimuthAngJ}, polarAngJ = ToSphericalCoordinates[Re[V]][[2]];
   azimuthAngJ = ToSphericalCoordinates[Re[V]][[3]];
   Return[{polarAngJ, azimuthAngJ}];];

vectorComponents[V_, θ_, φ_] := V * {Sin[θ] * Cos[φ], Sin[θ] * Sin[φ], Cos[θ]};

computeEulerMatrix[jvec_] :=
  Module[{θ, ϕ, cosθ, sinθ, cosϕ, sinϕ}, {θ, ϕ} = sphericalAngles[jvec] + {0, π / 2};
   cosθ = Cos[θ]; sinθ = Sin[θ];
   cosϕ = Cos[ϕ]; sinϕ = Sin[ϕ];
   N[{{cosϕ, sinϕ, 0},
     {-sinϕ * cosθ, cosϕ * cosθ, sinθ}, {sinϕ * sinθ, -cosϕ * sinθ, cosθ}}]];

(* ================================================================*)
(*BASE QUANTITIES COMPUTATION (t=0)*)
(* ================================================================*)

computeBaseQuantities2S[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, AVG_, ORB_] := Module[{
    o, c, M, μ, ν, δ, Δ, σ1, σ2, δ1, δ2,
    rvec, r, pvec, p, SS10, SS20,
    s1vec, s2vec, s1, s2, lvec, l, jvec, j, EulMat, InvEulMat,
    λ, lseff, mathcalSvec, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
    s2vecLframe0, ϕs1, ϕs2, s1p, s2p, a, b, ϕs, sp2, δl20, ltilde2, δl2avg,
    lavg, a0, e0, ar, er, et, eϕ, n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d,
    A3, A2, A1, A0, A, x2, x3, x1, β, SqA,
    α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1,
    drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosϰ10, Int30, sign,
    α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0,
    s1ξ10, solPieceS110, solPieceS120,
    JtoLframeEulMat0, r0Lframe, rϕ0, solPieceR10, solPieceR20,
    Nprec, Uprec, Tprec, Torb, freqcosϰ1, freqξ1L},
```

```
If[ORB == "Yes", o = 1];
If[ORB == "No", o = 0];

c = 1 / Sqrt[ϵ];
M = N[m1 + m2]; μ = N[m1 m2 / M]; ν = μ / M; δ = (m1 - m2) / M;
rvec = N[R0 / M]; r = Norm[rvec];
pvec = N[P0 / μ]; p = Norm[pvec];
SS10 = N[S10 Sqrt[ϵ]]; s1vec = SS10 / (μ M); s1 = Norm[s1vec];
SS20 = N[S20 Sqrt[ϵ]]; s2vec = SS20 / (μ M); s2 = Norm[s2vec];
lvec = Cross[rvec, pvec]; l = Norm[lvec];
jvec = lvec + s1vec + s2vec; j = Norm[jvec];
EulMat = computeEulerMatrix[jvec]; InvEulMat = Inverse[EulMat];

(*Spin-Coupling*)
δ1 = 2 ν (1 + 3 m2 / (4 m1));
δ2 = 2 ν (1 + 3 m1 / (4 m2));
Δ = δ1 + δ2 - ν / 2;
lseff = Dot[lvec, δ1 s1vec + δ2 s2vec];
σ1 = ν (1 + m2 / m1);
σ2 = ν (1 + m1 / m2);
λ = Dot[lvec, σ1 s1vec + σ2 s2vec] / l ^ 2;
mathcalSvec = (m2 / M) s1vec + (m1 / M) s2vec;
mathcalS2 = Norm[mathcalSvec] ^ 2;

(*h=hN+h1PN+hSO+h2PN+HSS*)h = p ^ 2 / 2 - 1 / r + o ϵ
    (1 / (2 r ^ 2) + 1 / 8 p ^ 4 (-1 + 3 ν) - (Dot[rvec / r, pvec] ^ 2 ν + p ^ 2 (3 + ν)) / (2 r)) +
   (lseff ϵ) / r ^ 3 + o ϵ ^ 2 (-((1 + 3 ν) / (4 r ^ 3)) + 1 / 16 p ^ 6 (1 - 5 ν + 5 ν ^ 2) +
      (3 Dot[rvec / r, pvec] ^ 2 ν + p ^ 2 (5 + 8 ν)) / (2 r ^ 2) + (-3 Dot[rvec / r, pvec] ^ 4
          ν ^ 2 - 2 Dot[rvec / r, pvec] ^ 2 p ^ 2 ν ^ 2 + p ^ 4 (5 - 20 ν - 3 ν ^ 2)) / (8 r)) -
   ((Dot[s1vec, s2vec] - 3 Dot[rvec / r, s1vec] × Dot[rvec / r, s2vec]) ϵ ν) / r ^ 3 +
   (m2 (3 Dot[rvec / r, s1vec] ^ 2 - s1 ^ 2) ϵ ν) / (2 m1 r ^ 3) +
   (m1 (3 Dot[rvec / r, s2vec] ^ 2 - s2 ^ 2) ϵ ν) / (2 m2 r ^ 3);
Σ1 = N[Dot[s1vec, s2vec] / (s1 s2) + l / s2 (m1 - m2) / m1 Dot[s1vec, lvec] / (l s1)];
Σ2 = N[Dot[s2vec, lvec] / (l s2) + (m2 s1) / (m1 s2) Dot[s1vec, lvec] / (l s1)];

(*Cubic equation coefficients*)
A3 = (2 l (m1 - m2) m2 s1) / m1 ^ 2;
A2 = - (1 / m1 ^ 2)
   (l ^ 2 (m1 - m2) ^ 2 + m2 ^ 2 s1 ^ 2 + m1 ^ 2 s2 ^ 2 + 2 m1 m2 s1 s2 Σ1 + 2 l m1 (m1 - m2) s2 Σ2);
A1 = (2 s2 (l (m1 - m2) Σ1 + (m2 s1 + m1 s2 Σ1) Σ2)) / m1;
A0 = s2 ^ 2 (1 - Σ1 ^ 2 - Σ2 ^ 2);
A = 9 / 2 (m2 (m1 - m2)) / M ^ 2 (1 - λ) ^ 2 l s1;
{x2, x3, x1} = Sort[x /. Solve[A3 x ^ 3 + A2 x ^ 2 + A1 x + A0 == 0, x]];
```

```
β = Sqrt[(x3 - x2) / (x1 - x2)];
SqA = 2 / Sqrt[A (x1 - x2)];
JtoLframeEulMat0 = computeEulerMatrix[EulMat.lvec];
r0Lframe = JtoLframeEulMat0.EulMat.rvec;
s1vecLframe0 = JtoLframeEulMat0.EulMat.s1vec;
s2vecLframe0 = JtoLframeEulMat0.EulMat.s2vec;

a0 = -(1 / (2 h)); e0 = Sqrt[1 + 2 h l^2];
ϕs1 = ArcTan[s1vecLframe0[[1]], s1vecLframe0[[2]]];
ϕs2 = ArcTan[s2vecLframe0[[1]], s2vecLframe0[[2]]];
s1p = Sqrt[s1vecLframe0[[1]]^2 + s1vecLframe0[[2]]^2];
s2p = Sqrt[s2vecLframe0[[1]]^2 + s2vecLframe0[[2]]^2];
a = m2/M s1p; b = m1/M s2p;
ϕs = ArcTan[a Cos[ϕs1] + b Cos[ϕs2], a Sin[ϕs1] + b Sin[ϕs2]];
sp2 = a^2 + b^2 + 2 a b Cos[ϕs1 - ϕs2];

δl20 = ϵ (2 ν h Dot[s1vec, s2vec] +
    2 / r (Dot[rvec / r, mathcalSvec]) ^ 2 - Dot[pvec, mathcalSvec] ^ 2);
ltilde2 = l ^ 2 + δl20;
δl2avg = ϵ (2 ν h Dot[s1vec, s2vec] + sp2 / (2 ltilde2) (1 - e0 ^ 2 + e0 ^ 2 Cos[2 ϕs]));
lavg = Sqrt[ltilde2 - δl2avg];

(*2PN QKP Orbital Elements*)
n = 2 Sqrt[2] (-h) ^ (3 / 2) *
  (1 + o 1 / 4 ϵ h (15 - ν) + o ϵ ^ 2 * (3 / 2 (-5 + 2 ν) (2 Sqrt[2] (-h) ^ (3 / 2)) / lavg +
      1 / 32 (555 + 30 ν + 11 ν ^ 2) h ^ 2));
nunscaled = n / M;
ar =
 -1 / (2 h) * (1 + o 1 / 2 ϵ h (7 - ν) - 1 / 2 ϵ h / lavg ^ 2 (4 lseff - 2 mathcalS2 + 3 sp2) +
    o 1 / 4 ϵ ^ 2 (2 (17 - 11 ν) h / lavg ^ 2 + (1 + 10 ν + ν ^ 2) h ^ 2));
er = Sqrt[1 + 2 h lavg ^ 2 - o ϵ (2 (6 - ν) h + 5 (3 - ν) h ^ 2 lavg ^ 2) +
   ϵ 2 h / lavg ^ 2 (1 + h lavg ^ 2) (4 lseff - 2 mathcalS2 + 3 sp2) + o ϵ ^ 2
    (-2 (17 - 11 ν) h / lavg ^ 2 + (26 + ν + ν ^ 2) h ^ 2 + (80 - 55 ν + 4 ν ^ 2) h ^ 3 lavg ^ 2)];
et = Sqrt[1 + 2 h lavg ^ 2 + o ϵ ( 4 (1 - ν) h + (17 - 7 ν) h ^ 2 lavg ^ 2) +
   ϵ h / lavg ^ 2 (4 lseff - 2 mathcalS2 + 3 sp2) + o ϵ ^ 2
    ((11 ν - 17) h / lavg ^ 2 + 2 (2 + ν + 5 ν ^ 2) h ^ 2 + (112 - 47 ν + 16 ν ^ 2) h ^ 3 lavg ^ 2 +
      3 (2 ν - 5) (1 + 2 h lavg ^ 2) (2 Sqrt[2] (-h) ^ (3 / 2)) / lavg)];
eϕ = o Sqrt[1 + 2 h lavg ^ 2 + ϵ h (-( 12 - 4 Δ) - ( 15 - ν - 8 Δ) h lavg ^ 2 +
        1 / lavg ^ 2 (3 + 4 h lavg ^ 2) (4 lseff - 2 mathcalS2 + 3 sp2) -
        2 (1 + 2 h lavg ^ 2) λ ((2 Δ - ν) + 3 s2 / lavg μ / m2 Σ2)) +
     ϵ ^ 2 (1 / 8 (-408 + 232 ν + 15 ν ^ 2 + 64 Δ (4 - 5 / 2 ν + Δ)) h / lavg ^ 2 -
        1 / 2 (16 - 88 ν - 9 ν ^ 2 - 12 Δ (1 - 9 ν + 14 / 3 Δ)) h ^ 2 +
```

```
        1 / 2 (160 - 30 ν + 3 ν ^ 2 - 16 Δ (10 + 3 ν - 3 Δ)) h ^ 3 lavg ^ 2)] +
  (1 - o) Sqrt[1 + 2 h lavg ^ 2 + 4 ϵ h (1 + 2 h lavg ^ 2) Δ +
     h (64 + 224 h lavg^2 + 192 h^2 lavg^4) Δ^2 ϵ^2 / (8 lavg^2) +
     ϵ (-4 h (-3 - 4 h lavg^2) (lseff / lavg ^ 2 + (3 sp2 - 2 mathcalS2) / lavg ^ 2 / 4 ) -
        2 h (1 + 2 h lavg^2) λ (-ν + 2 Δ + 3 m1 s2 Σ2 / lavg / M))];
ft = - o 1 / 8 ν (4 + ν) Sqrt[1 + 2 h lavg ^ 2] (2 Sqrt[2] (-h) ^ (3 / 2)) / lavg;
gt = o 3 / 2 (5 - 2 ν) (2 Sqrt[2] (-h) ^ (3 / 2)) / lavg;
fϕ = o 1 / 8 (ν - 3 ν ^ 2 - Δ (8 - 5 ν + 2 Δ)) (1 + 2 h lavg ^ 2) / lavg ^ 4 -
  (1 - o) Δ^2 / 4 (1 + 2 h lavg ^ 2) / lavg ^ 4;
gϕ = -o 3 ν ^ 2 / 32 (1 + 2 h lavg ^ 2) ^ (3 / 2) / lavg ^ 4;
Φ = o (1 + ϵ / lavg ^ 2 * ( 3 - Δ - 3 / (4 lavg ^ 2) (4 lseff - 2 mathcalS2 + 3 sp2) +
        1 / 2 λ ((2 Δ - ν) + 3 s2 / lavg μ / m2 Σ2) + ϵ / (4 lavg ^ 2)
         (105 - 30 ν - 60 Δ + 15 Δ ν - 2 h lavg ^ 2 (-15 + 6 ν + Δ (18 - 13 ν))))) +
  (1 - o) (1 + ϵ / lavg ^ 2 * ( -Δ - 3 / (4 lavg ^ 2) (4 lseff - 2 mathcalS2 + 3 sp2) +
        1 / 2 λ ((2 Δ - ν) + 3 s2 / lavg μ / m2 Σ2)));
hr = -sp2 / (4 lavg ^ 2);
hϕ1 = -1 / 2 Sqrt[1 + 2 h lavg ^ 2] sp2 / lavg ^ 4;
hϕ2 = -sp2 / (8 lavg ^ 4);
d = ((2 + er ^ 2 - 3 er et) / (2 ar ^ 3 (1 - er ^ 2) ^ (5 / 2))) ^ (-1 / 3);

(*Alphas and Betas*)
α1l = m1 / (m1 - m2) (j + l + s2 Σ2) / s1;
α2l = m1 / (m1 - m2) (-j + l + s2 Σ2) / s1;
β1l = (3 m1) / (4 (m1 ^ 2 - m2 ^ 2)) 1 / s1 (1 - λ)
  (m1 j ^ 2 + m2 l ^ 2 + (m1 + m2) j l - (m1 - m2) s1 (s1 + s2 Σ1) + (m2 l + m1 j) s2 Σ2);
β2l = (3 m1) / (4 (m1 ^ 2 - m2 ^ 2)) 1 / s1 (1 - λ)
  (m1 j ^ 2 + m2 l ^ 2 - (m1 + m2) j l - (m1 - m2) s1 (s1 + s2 Σ1) + (m2 l - m1 j) s2 Σ2);
β3l = (j ν) / 2;

α1s1 = m1 / m2 1 / l (-j + s1 + s2 Σ1);
α2s1 = m1 / m2 1 / l (j + s1 + s2 Σ1);
β1s1 = m1 / (4 m2 l) ((2 δ2 - 3 μ / m2 λ) j ^ 2 - (2 δ1 - 3 μ / m1 λ) l ^ 2 -
    ν (s1 ^ 2 + s2 ^ 2 + 2 s1 s2 Σ1 + 2 l s2 Σ2) - 3 ((2 m1 - m2) / (m1 + m2)) (1 - λ) j s1 +
    3 (m1 ^ 2 - m2 ^ 2) / (m1 + m2) ^ 2 (1 - λ) (s1 ^ 2 + s1 s2 Σ1) -
    3 (1 - λ) / (m1 + m2) (m1 j Σ1 + m2 l Σ2) s2);
β2s1 = m1 / (4 m2 l) ((2 δ2 - 3 μ / m2 λ) j ^ 2 - (2 δ1 - 3 μ / m1 λ) l ^ 2 -
    ν (s1 ^ 2 + s2 ^ 2 + 2 s1 s2 Σ1 + 2 l s2 Σ2) + 3 ((2 m1 - m2) / (m1 + m2)) (1 - λ) j s1 +
    3 (m1 ^ 2 - m2 ^ 2) / (m1 + m2) ^ 2 (1 - λ) (s1 ^ 2 + s1 s2 Σ1) -
    3 (1 - λ) / (m1 + m2) (-m1 j Σ1 + m2 l Σ2) s2);
β3s1 = (δ2 - 3 μ / (2 m2) λ) j;
```

```
(*Compute u0, t0 and ϒ0*)
βeϕ = eϕ / (1 + Sqrt[1 - eϕ^2]);
βer = er / (1 + (1 - er^2)^(1/2));
drdt0 = Sign[pvec.rvec / r + o ϵ (-6 - 4 ν + p^2 r (-1 + 3 ν)) pvec.rvec / 2 r^2 +
    o ϵ^2 (r (40 + 88 ν - 4 p^2 r (-5 + 20 ν + 4 ν^2) + 3 p^4 r^2 (1 - 5 ν + 5 ν^2)) pvec.rvec -
        16 o ν^2 (pvec.rvec)^3) / 8 r^4];
u0test = ArcCos[Clip[(r / ar - 1) / (-er), {-1, 1}]];
Absu0 = u /. FindRoot[r == ar (1 - er Cos[u]) + ϵ hr
       Cos[2 (u + 2 ArcTan[(βeϕ Sin[u]) / (1 - βeϕ Cos[u])]) - 2 * ϕs], {u, u0test}];
u0 = drdt0 * Absu0;
veϕ0 = u0 + 2 ArcTan[(βeϕ Sin[u0]) / (1 - βeϕ Cos[u0])];
ver0 = u0 + 2 ArcTan[(βer Sin[u0]) / (1 - βer Cos[u0])];
t0 = - (u0 - et Sin[u0] + ϵ^2 (ft Sin[veϕ0] + gt (veϕ0 - u0))) / nunscaled;
sign = If[(M - λ μ) Dot[lvec, Cross[s1vec, s2vec]] > 0, 1, -1];
cosϰ10 = Dot[lvec, s1vec] / (l s1);

If[AVG == "Yes",
 Int30 = (2 + er^2 - 3 er et) / (2 ar^3 (1 - er^2)^(5/2) n) (u0 - et Sin[u0]),
 Int30 = (1 / (2 ar^3 n)) * (((2 + er^2 - 3 er et) ver0) / (1 - er^2)^(5/2) +
      ((-er + et) Sin[u0]) / ((-1 + er^2) (-1 + er Cos[u0])^2) +
      ((-3 er + et + 2 er^2 et) Sin[u0]) / ((-1 + er^2)^2 (-1 + er Cos[u0])))];
α = sign * 2 / Sqrt[A (x1 - x2)] *
   EllipticF[ArcSin[Sqrt[(cosϰ10 - x2) / (x3 - x2)]], β^2] - (ϵ * Int30);
ϒ0 = 0.5 Sqrt[A (x1 - x2)] (α + ϵ Int30);

(*SolPieces mappings for L,S1,R*)
lvecJframe0 = EulMat.lvec;
lξ10 = sphericalAngles[lvecJframe0][[2]] + π / 2;
solPieceL10 =
 SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), JacobiAmplitude[ϒ0, β^2], β^2] /
       (α1l + x2) - β2l EllipticPi[(x2 - x3) / (α2l + x2),
        JacobiAmplitude[ϒ0, β^2], β^2] / (α2l + x2) + β3l ϒ0);
solPieceL20 = lξ10 - solPieceL10;

s1vecJframe0 = EulMat.s1vec;
s1ξ10 = sphericalAngles[s1vecJframe0][[2]] + π / 2;
solPieceS110 =
 SqA (β1s1 EllipticPi[(x2 - x3) / (α1s1 + x2), JacobiAmplitude[ϒ0, β^2], β^2] /
       (α1s1 + x2) - β2s1 EllipticPi[(x2 - x3) / (α2s1 + x2),
        JacobiAmplitude[ϒ0, β^2], β^2] / (α2s1 + x2) + β3s1 ϒ0);
solPieceS120 = s1ξ10 - solPieceS110;

r0Lframe = JtoLframeEulMat0.EulMat.rvec;
```

```
    rϕ0 = sphericalAngles[r0Lframe][[2]];
    solPieceR10 =
     SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), JacobiAmplitude[ϒ0, β^2], β^2] /
           (α1l + x2) + β2l EllipticPi[(x2 - x3) / (α2l + x2),
            JacobiAmplitude[ϒ0, β^2], β^2] / (α2l + x2));
    solPieceR20 = Φ veϕ0 + ϵ^2 (fϕ Sin[2 veϕ0] + gϕ Sin[3 veϕ0]) +
      ϵ (hϕ1 Sin[veϕ0 - 2 ϕs] + hϕ2 Sin[2 veϕ0 - 2 ϕs]);

    (*Frequencies*)
    Nprec = Hypergeometric2F1[1 / 2, 1 / 2, 1, β^2]
      (2 ar^3 n (1 - er^2)^(5 / 2)) / (ϵ Sqrt[A (x1 - x2)] (2 + er^2 - 3 et er));
    Uprec = 2 π Nprec;
    Tprec = Uprec / n M;
    Torb = 2 π / nunscaled;
    freqξ1L = (ϵ ((β1l EllipticPi[(x2 - x3) / (x2 + α1l), β^2]) / (x2 + α1l) -
           (β2l EllipticPi[(x2 - x3) / (x2 + α2l), β^2]) / (x2 + α2l) +
          β3l EllipticK[β^2])) / (d^3 M EllipticK[β^2]);
    freqcosκ1 = nunscaled / Nprec;
    Return[N[{c, M, μ, ν, rvec, r, pvec, p, SS10, SS20, s1vec, s2vec, s1, s2, lvec, l,
       jvec, j, σ1, σ2, λ, δ1, δ2, δ, Δ, lseff, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
       s2vecLframe0, δl20, ltilde2, δl2avg, lavg, a0, e0, ϕs, sp2, ar, er, et, eϕ,
       n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d, A3, A2, A1, A0, A, x2, x3,
       x1, β, SqA, α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1, EulMat,
       InvEulMat, drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosκ10, Int30,
       sign, α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0,
       s1ξ10, solPieceS110, solPieceS120, JtoLframeEulMat0, r0Lframe, rϕ0,
       solPieceR10, solPieceR20, Nprec, Uprec, Tprec, Torb, freqcosκ1, freqξ1L}]];];
```

```
(*==*)(*TIME PROPAGATION MODULE (2S)*)(*==*)

Hflow2SList[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, AVG_, ORB_, tList_] :=
  Module[{pi, c, M, μ, ν, rvec, r, pvec, p, SS10, SS20, s1vec, s2vec, s1, s2, lvec, l,
    jvec, j, σ1, σ2, λ, δ1, δ2, δ, Δ, lseff, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
    s2vecLframe0, δl20, ltilde2, δl2avg, lavg, a0, e0, spVecX, spVecY, ϕs, sp2, ar,
    er, et, eϕ, n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d, A3, A2, A1, A0, A,
    x2, x3, x1, β, SqA, α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1, EulMat,
    InvEulMat, drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosκ10, Int30,
    sign, α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0, s1ξ10,
    solPieceS110, solPieceS120, JtoLframeEulMat0, r0Lframe, rϕ0, solPieceR10,
```

```
 solPieceR20, Nprec, Uprec, Tprec, Torb, freqcosκ1, freqξ1L, computeForTime},

pi = 3.141592653589793`;
{c, M, μ, ν, rvec, r, pvec, p, SS10, SS20, s1vec, s2vec, s1, s2, lvec, l, jvec,
  j, σ1, σ2, λ, δ1, δ2, δ, Δ, lseff, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
  s2vecLframe0, δl20, ltilde2, δl2avg, lavg, a0, e0, ϕs, sp2, ar, er, et, eϕ,
  n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d, A3, A2, A1, A0, A, x2, x3,
  x1, β, SqA, α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1, EulMat,
  InvEulMat, drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosκ10, Int30,
  sign, α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0,
  s1ξ10, solPieceS110, solPieceS120, JtoLframeEulMat0, r0Lframe, rϕ0,
  solPieceR10, solPieceR20, Nprec, Uprec, Tprec, Torb, freqcosκ1, freqξ1L} =
 computeBaseQuantities2S[m1, m2, R0, P0, S10, S20, ϵ, AVG, ORB];

(*Evolution for time t*)
computeForTime[tmax_] :=
 Module[{o, fveϕ, findu, umax, cosu, sinu, veϕ, ver, Int3, ϒ, JacAmp, cosκ1,
   solPieceL1, lξ1sol, cosκ2,
   lvecAzimuthAngle, lvecPolarAngle, LsolJframe, Lsol,
   solPieceS11, s1ξ1sol, cosγ,
   s1vecAzimuthAngle, s1vecPolarAngle, S1sol, S2sol,
   JtoLframeEulMat, s1vecLframe, s2vecLframe,
   Uϕs1, Uϕs2, Us1p, Us2p, Ua, Ub, Uϕs, Usp2, Us1vec, Us2vec,
   Ulvec, Ulseff, Uλ, UmathcalSvec, UmathcalS2, Uδl2avg, Ulavg,
   Un, Ununscaled, Uar, Uer, Uet, Ueϕ, Uft,
   Ugt, Ufϕ, Ugϕ, UΦ, Uhr, Uhϕ1, Uhϕ2, Uumax, Usinu, Ucosu, Uveϕ,
   Uβeϕ, Ufveϕ, Ufindu, Uβer, Uver, UInt3, Uϒ, UJacAmp, solPieceR1, solPieceR2,
   solPieceR3, solPieceR4, rvecAzimuthAngleLframe, rMagnitude, Rsol},

  If[ORB == "Yes", o = 1];
  If[ORB == "No", o = 0];

  βeϕ = eϕ / (1 + Sqrt[1 - eϕ^2]);
  fveϕ = u + 2. ArcTan[(βeϕ Sin[u]) / (1 - βeϕ Cos[u])];

  findu[t_] := Module[{target, initialGuess}, target = nunscaled (t - t0);
    initialGuess = target;
    FindRoot[target == u - et Sin[u] + ϵ^2 (ft Sin[fveϕ] + gt (fveϕ - u)), {u,
      initialGuess}, Method → "Newton", WorkingPrecision → MachinePrecision]];

  umax = u /. findu[tmax];
  cosu = Cos[umax]; sinu = Sin[umax];
  veϕ = umax + 2. ArcTan[(βeϕ sinu) / (1 - βeϕ cosu)];
```

```
ver = umax + 2. ArcTan[(βer sinu) / (1 - βer cosu)];

If[AVG == "Yes",
 Int3 = (2 + er^2 - 3 er et) / (2 ar^3 (1 - er^2)^(5/2) n) (umax - et sinu),
 Int3 = (1. / (2. ar^3. n)) (((2. + er^2. - 3. er et) ver) / (1. - er^2.)^(5./2.) +
      ((-er + et) sinu) / ((-1. + er^2.) (-1. + er cosu)^2.) +
      ((-3. er + et + 2. er^2. et) sinu) / ((-1. + er^2.)^2. (-1. + er cosu)))];
ϒ = 0.5 Sqrt[A (x1 - x2)] (α + ϵ Int3);
JacAmp = JacobiAmplitude[ϒ, β^2];
cosκ1 = x2 + (x3 - x2) JacobiSN[ϒ, β^2]^2;

(*L Vector Evolution*)
solPieceL1 = SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), JacAmp, β^2] / (α1l + x2)
    - β2l EllipticPi[(x2 - x3) / (α2l + x2), JacAmp, β^2] / (α2l + x2)
    + β3l ϒ);
lξ1sol = solPieceL1 + solPieceL20;
cosκ2 = Σ2 - m2 s1 / (m1 s2) cosκ1;
lvecAzimuthAngle = -pi / 2. + lξ1sol;
lvecPolarAngle = ArcCos[(l^2 + l s1 cosκ1 + l s2 cosκ2) / (l j)];
LsolJframe = μ M vectorComponents[l, lvecPolarAngle, lvecAzimuthAngle];
Lsol = (InvEulMat.LsolJframe);

(*S1 Vector Evolution*)
solPieceS11 =
 SqA (β1s1 EllipticPi[(x2 - x3) / (α1s1 + x2), JacAmp, β^2] / (α1s1 + x2)
    - β2s1 EllipticPi[(x2 - x3) / (α2s1 + x2), JacAmp, β^2] / (α2s1 + x2)
    + β3s1 ϒ);
s1ξ1sol = solPieceS11 + solPieceS120;
cosγ = Σ1 - l (m1 - m2) / (m1 s2) cosκ1;
s1vecAzimuthAngle = -pi / 2. + s1ξ1sol;
s1vecPolarAngle = ArcCos[(s1^2. + l s1 cosκ1 + s1 s2 cosγ) / (s1 j)];
S1sol =
 μ M (InvEulMat.vectorComponents[s1, s1vecPolarAngle, s1vecAzimuthAngle]);

(*S2 Vector Evolution*)
S2sol = (jvec * μ * M) - Lsol - S1sol;

(*Dynamic "U"pdating for R Vector*)
JtoLframeEulMat = computeEulerMatrix[LsolJframe];
s1vecLframe = JtoLframeEulMat.EulMat.S1sol / (μ M);
s2vecLframe = JtoLframeEulMat.EulMat.S2sol / (μ M);

a0 = -(1 / (2 h)); e0 = Sqrt[1 + 2 h l^2];
Uϕs1 = ArcTan[s1vecLframe[[1]], s1vecLframe[[2]]];
```

```
Uϕs2 = ArcTan[s2vecLframe⟦1⟧, s2vecLframe⟦2⟧];
Us1p = Sqrt[s1vecLframe⟦1⟧^2 + s1vecLframe⟦2⟧^2];
Us2p = Sqrt[s2vecLframe⟦1⟧^2 + s2vecLframe⟦2⟧^2];
Ua = m2/M Us1p; Ub = m1/M Us2p;
Uϕs = ArcTan[Ua Cos[Uϕs1] + Ub Cos[Uϕs2], Ua Sin[Uϕs1] + Ub Sin[Uϕs2]];
Usp2 = Ua^2 + Ub^2 + 2 Ua Ub Cos[Uϕs1 - Uϕs2];

Us1vec = S1sol / (μ M); Us2vec = S2sol / (μ M); Ulvec = Lsol / (μ M);
Ulseff = Dot[Ulvec, δ1 Us1vec + δ2 Us2vec];
Uλ = Dot[Ulvec, σ1 Us1vec + σ2 Us2vec] / Dot[Ulvec, Ulvec];
UmathcalSvec = (m2 / M) Us1vec + (m1 / M) Us2vec;
UmathcalS2 = Norm[UmathcalSvec]^2;
Uδl2avg =
 ϵ (2 ν h Dot[Us1vec, Us2vec] + Usp2 / (2 ltilde2) (1 - e0^2 + e0^2 Cos[2 Uϕs]));
Ulavg = Sqrt[ltilde2 - Uδl2avg];
(*Ulavg=lavg;*)

Un = 2 Sqrt[2] (-h)^(3/2) * (1 + o 1/4 ϵ h (15 - ν) + o ϵ^2 * (3/2 (-5 + 2 ν)
        (2 Sqrt[2] (-h)^(3/2)) / Ulavg + 1/32 (555 + 30 ν + 11 ν^2) h^2));
Ununscaled = Un / M;
Uar = -1/(2 h) *
  (1 + o 1/2 ϵ h (7 - ν) - 1/2 ϵ h / Ulavg^2 (4 Ulseff - 2 UmathcalS2 + 3 Usp2) +
    o 1/4 ϵ^2 (2 (17 - 11 ν) h / Ulavg^2 + (1 + 10 ν + ν^2) h^2));
Uer = Sqrt[1 + 2 h Ulavg^2 - o ϵ (2 (6 - ν) h + 5 (3 - ν) h^2 Ulavg^2) +
   ϵ 2 h / Ulavg^2 (1 + h Ulavg^2) (4 Ulseff - 2 UmathcalS2 + 3 Usp2) +
   o ϵ^2 (-2 (17 - 11 ν) h / Ulavg^2 + (26 + ν + ν^2) h^2 +
      (80 - 55 ν + 4 ν^2) h^3 Ulavg^2)];
Uet = Sqrt[1 + 2 h Ulavg^2 + o ϵ ( 4 (1 - ν) h + (17 - 7 ν) h^2 Ulavg^2) +
   ϵ h / Ulavg^2 (4 Ulseff - 2 UmathcalS2 + 3 Usp2) +
   o ϵ^2 ((11 ν - 17) h / Ulavg^2 + 2 (2 + ν + 5 ν^2) h^2 + (112 - 47 ν + 16 ν^2) h^3
       Ulavg^2 + 3 (2 ν - 5) (1 + 2 h Ulavg^2) (2 Sqrt[2] (-h)^(3/2)) / Ulavg)];
Ueϕ = o Sqrt[1 + 2 h Ulavg^2 + ϵ h (-( 12 - 4 Δ) - ( 15 - ν - 8 Δ) h Ulavg^2 +
         1 / Ulavg^2 (3 + 4 h Ulavg^2) (4 Ulseff - 2 UmathcalS2 + 3 Usp2) -
         2 (1 + 2 h Ulavg^2) Uλ ((2 Δ - ν) + 3 s2 / Ulavg μ / m2 Σ2)) +
      ϵ^2 (1/8 (-408 + 232 ν + 15 ν^2 + 64 Δ (4 - 5/2 ν + Δ)) h / Ulavg^2 -
         1/2 (16 - 88 ν - 9 ν^2 - 12 Δ (1 - 9 ν + 14/3 Δ)) h^2 +
         1/2 (160 - 30 ν + 3 ν^2 - 16 Δ (10 + 3 ν - 3 Δ)) h^3 Ulavg^2)] +
  (1 - o) Sqrt[1 + 2 h Ulavg^2 + 4 ϵ h (1 + 2 h Ulavg^2) Δ +
     h (64 + 224 h Ulavg^2 + 192 h^2 Ulavg^4) Δ^2 ϵ^2 / (8 Ulavg^2) +
     ϵ (-4 h (-3 - 4 h Ulavg^2) (Ulseff / Ulavg^2 + (3 Usp2 - 2 UmathcalS2) / Ulavg^2 /
```

```
            4 ) - 2 h (1 + 2 h Ulavg^2) Uλ (-ν + 2 Δ + 3 m1 s2 Σ2 / Ulavg / M))];
Uft = - o 1 / 8 ν (4 + ν) Sqrt[1 + 2 h Ulavg^2] (2 Sqrt[2] (-h)^(3 / 2)) / Ulavg;
Ugt = o 3 / 2 (5 - 2 ν) (2 Sqrt[2] (-h)^(3 / 2)) / Ulavg;
Ufϕ = o 1 / 8 (ν - 3 ν^2 - Δ (8 - 5 ν + 2 Δ)) (1 + 2 h Ulavg^2) / Ulavg^4 -
  (1 - o) Δ^2 / 4 (1 + 2 h Ulavg^2) / Ulavg^4;
Ugϕ = -o 3 ν^2 / 32 (1 + 2 h Ulavg^2)^(3 / 2) / Ulavg^4;
UΦ = o (1 + ϵ / Ulavg^2 * ( 3 - Δ - 3 / (4 Ulavg^2) (4 Ulseff - 2 UmathcalS2 + 3 Usp2) +
        1 / 2 Uλ ((2 Δ - ν) + 3 s2 / Ulavg μ / m2 Σ2) + ϵ / (4 Ulavg^2)
          (105 - 30 ν - 60 Δ + 15 Δ ν - 2 h Ulavg^2 (-15 + 6 ν + Δ (18 - 13 ν))))) +
  (1 - o) (1 + ϵ / Ulavg^2 * ( -Δ - 3 / (4 Ulavg^2) (4 Ulseff - 2 UmathcalS2 + 3 Usp2) +
        1 / 2 Uλ ((2 Δ - ν) + 3 s2 / Ulavg μ / m2 Σ2)));


Uhr = -Usp2 / (4 Ulavg^2);
Uhϕ1 = -1 / 2 Sqrt[1 + 2 h Ulavg^2] Usp2 / Ulavg^4;
Uhϕ2 = -Usp2 / (8 Ulavg^4);

Uβeϕ = Ueϕ / (1 + Sqrt[1 - Ueϕ^2]);
Ufveϕ = u + 2. ArcTan[(Uβeϕ Sin[u]) / (1 - Uβeϕ Cos[u])];

Ufindu[t_] := Module[{target, initialGuess}, target = Ununscaled (t - t0);
  initialGuess = target;
  FindRoot[target == u - Uet Sin[u] + ϵ^2 (Uft Sin[Ufveϕ] + Ugt (Ufveϕ - u)), {u,
    initialGuess}, Method → "Newton", WorkingPrecision → MachinePrecision]];
Uumax = u /. Ufindu[tmax];
Usinu = Sin[Uumax]; Ucosu = Cos[Uumax];
Uveϕ = Uumax + 2. ArcTan[(Uβeϕ Usinu) / (1 - Uβeϕ Ucosu)];
Uβer = Uer / (1 + Sqrt[1 - Uer^2]);
Uver = Uumax + 2. ArcTan[(Uβer Usinu) / (1 - Uβer Ucosu)];

If[AVG == "Yes", UInt3 =
  (2 + Uer^2 - 3 Uer Uet) / (2 Uar^3 (1 - Uer^2)^(5 / 2) Un) (Uumax - Uet Usinu),
 UInt3 = (1. / (2. Uar^3. Un)) (((2. + Uer^2. - 3. Uer Uet) Uver) /
      (1. - Uer^2.)^(5. / 2.) + ((-Uer + Uet) Usinu) /
      ((-1. + Uer^2.) (-1. + Uer Ucosu)^2.) + ((-3. Uer + Uet + 2. Uer^2. Uet)
        Usinu) / ((-1. + Uer^2.)^2. (-1. + Uer Ucosu)))];
UΥ = 0.5 Sqrt[A (x1 - x2)] (α + ϵ UInt3);
UJacAmp = JacobiAmplitude[UΥ, β^2];

solPieceR1 =  SqA (β1l EllipticPi[(x2 - x3) / (α1l + x2), UJacAmp, β^2] / (α1l + x2)
    + β2l EllipticPi[(x2 - x3) / (α2l + x2), UJacAmp, β^2] / (α2l + x2));
solPieceR2 = UΦ Uveϕ + (ϵ^2 (Ufϕ Sin[2 Uveϕ] + Ugϕ Sin[3 Uveϕ]) +
    ϵ (Uhϕ1 Sin[Uveϕ - 2 Uϕs] + Uhϕ2 Sin[2 Uveϕ - 2 Uϕs]));
```

```
    solPieceR3 = solPieceR1 + solPieceR2;
    solPieceR4 = rϕ0 - (solPieceR10 + solPieceR20);
    rvecAzimuthAngleLframe = solPieceR3 + solPieceR4;
    (*rvecAzimuthAngleLframe=AzimuthalRvectorLframe[tmax,Num2S,data][[1]];*)

    rMagnitude = Uar (1 - Uer Ucosu) + ϵ Uhr Cos[2 Uveϕ - 2 Uϕs];

    Rsol = M (InvEulMat.Transpose[JtoLframeEulMat].
        vectorComponents[rMagnitude, Pi / 2, rvecAzimuthAngleLframe]);
    {(*1*)Rsol, (*2*)Lsol, (*3*)S1sol, (*4*)S2sol, (*5*)Tprec, (*6*)
     Torb, (*7*)rvecAzimuthAngleLframe, (*8*)Transpose[JtoLframeEulMat],
     (*9*)vectorComponents[rMagnitude, Pi / 2, rvecAzimuthAngleLframe]}];

  Map[computeForTime, tList]];
```

C. HN + H1PN + H2PN : Fixed Spins

```
In[∘]:= computeBaseQuantitiesFixedSpin[m1_,
    m2_, R0_, P0_, S10_, S20_, ϵ_, AVG_, ORB_] := Module[{
     o, c, M, μ, ν, δ, Δ, σ1, σ2, δ1, δ2,
     rvec, r, pvec, p, SS10, SS20,
     s1vec, s2vec, s1, s2, lvec, l, jvec, j, EulMat, InvEulMat,
     λ, lseff, mathcalSvec, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
     s2vecLframe0, ϕs1, ϕs2, s1p, s2p, a, b, ϕs, sp2, δl20, ltilde2, δl2avg,
     lavg, a0, e0, ar, er, et, eϕ, n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d,
     A3, A2, A1, A0, A, x2, x3, x1, β, SqA,
     α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1,
     drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosκ10, Int30, sign,
     α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0,
     s1ξ10, solPieceS110, solPieceS120,
     JtoLframeEulMat0, r0Lframe, rϕ0, solPieceR10, solPieceR20,
     Nprec, Uprec, Tprec, Torb, freqcosκ1, freqξ1L},

    If[ORB == "Yes", o = 1];
    If[ORB == "No", o = 0];

    c = 1 / Sqrt[ϵ];
    M = N[m1 + m2]; μ = N[m1 m2 / M]; ν = μ / M; δ = (m1 - m2) / M;
    rvec = N[R0 / M]; r = Norm[rvec];
    pvec = N[P0 / μ]; p = Norm[pvec];
    s1vec = {0., 0., 0.};
```

```
s2vec = {0., 0., 0.};
s1 = 0.;
s2 = 0.;

lvec = Cross[rvec, pvec]; l = Norm[lvec];

EulMat = computeEulerMatrix[lvec]; InvEulMat = Inverse[EulMat];

(*h=hN+h1PN+hSO+h2PN+HSS*)
h = p^2 / 2 - 1 / r + o ϵ
    (1 / (2 r^2) + 1 / 8 p^4 (-1 + 3 ν) - (Dot[rvec / r, pvec]^2 ν + p^2 (3 + ν)) / (2 r)) +
  o ϵ^2 (-((1 + 3 ν) / (4 r^3)) + 1 / 16 p^6 (1 - 5 ν + 5 ν^2) +
      (3 Dot[rvec / r, pvec]^2 ν + p^2 (5 + 8 ν)) / (2 r^2) + (-3 Dot[rvec / r, pvec]^4
          ν^2 - 2 Dot[rvec / r, pvec]^2 p^2 ν^2 + p^4 (5 - 20 ν - 3 ν^2)) / (8 r));

r0Lframe = EulMat.rvec;


(*2PN QKP Orbital Elements*)
n = 2 Sqrt[2] (-h)^(3 / 2) * (1 + o 1 / 4 ϵ h (15 - ν) + o ϵ^2 *
      (3 / 2 (-5 + 2 ν) (2 Sqrt[2] (-h)^(3 / 2)) / l + 1 / 32 (555 + 30 ν + 11 ν^2) h^2)) ;
nunscaled = n / M;
ar =  -1 / (2 h) *
   (1 + o 1 / 2 ϵ h (7 - ν) + o 1 / 4 ϵ^2 (2 (17 - 11 ν) h / l^2 + (1 + 10 ν + ν^2) h^2));
er = Sqrt[1 + 2 h l^2 - o ϵ (2 (6 - ν) h + 5 (3 - ν) h^2 l^2) +
    o ϵ^2 (-2 (17 - 11 ν) h / l^2 + (26 + ν + ν^2) h^2 + (80 - 55 ν + 4 ν^2) h^3 l^2)];
et = Sqrt[1 + 2 h l^2 + o ϵ (4 (1 - ν) h + (17 - 7 ν) h^2 l^2) +
    o ϵ^2 ((11 ν - 17) h / l^2 + 2 (2 + ν + 5 ν^2) h^2 + (112 - 47 ν + 16 ν^2) h^3 l^2 +
       3 (2 ν - 5) (1 + 2 h l^2) (2 Sqrt[2] (-h)^(3 / 2)) / l)];
eϕ = 1 + 2 h l^2 + o (+ 12 (-h) ϵ - (15 - ν) h^2 l^2 ϵ - 1/2 (16 - 88 ν - 9 ν^2) h^2 ϵ^2 -
       1/2 (160 - 30 ν + 3 ν^2) (-h)^3 l^2 ϵ^2 + 1/8 (408 - 232 ν - 15 ν^2) (-h)/l^2 ϵ^2 ) // Sqrt ;
ft = -o 1 / 8 ν (4 + ν) Sqrt[1 + 2 h l^2] (2 Sqrt[2] (-h)^(3 / 2)) / l;
gt = o 3 / 2 (5 - 2 ν) (2 Sqrt[2] (-h)^(3 / 2)) / l;
fϕ = o 1 / 8 (ν - 3 ν^2) (1 + 2 h l^2) / l^4 ;
gϕ = -o 3 ν^2 / 32 (1 + 2 h l^2)^(3 / 2) / l^4;
Φ = o (1 + ϵ / l^2 * (3 + ϵ / (4 l^2) (105 - 30 ν - 2 h l^2 (-15 + 6 ν)))) + (1 - o);

(*============================*)

(*Compute u0, t0 and ϒ0*)
βeϕ = eϕ / (1 + Sqrt[1 - eϕ^2]);
```

```
βer = er / (1 + (1 - er ^ 2) ^ (1 / 2));
drdt0 = Sign[pvec.rvec / r + o ϵ (-6 - 4 ν + p^2 r (-1 + 3 ν)) pvec.rvec / 2 r^2 +
   o ϵ^2 (r (40 + 88 ν - 4 p^2 r (-5 + 20 ν + 4 ν^2) + 3 p^4 r^2 (1 - 5 ν + 5 ν^2)) pvec.rvec -
       16 o ν^2 (pvec.rvec)^3) / 8 r^4];
u0test = ArcCos[Clip[(r / ar - 1) / (-er), {-1, 1}]];
Absu0 = u /. FindRoot[r == ar (1 - er Cos[u]) (*+ϵ hr
    Cos[2(u+2 ArcTan[(βeϕ Sin[u])/(1-βeϕ Cos[u])])-2*ϕs ]*), {u, u0test}];
u0 = drdt0 * Absu0;
veϕ0 = u0 + 2 ArcTan[(βeϕ Sin[u0]) / (1 - βeϕ Cos[u0])];
ver0 = u0 + 2 ArcTan[(βer Sin[u0]) / (1 - βer Cos[u0])];
t0 = - (u0 - et Sin[u0] + ϵ ^ 2 (ft Sin[veϕ0] + gt (veϕ0 - u0))) / nunscaled;


rϕ0 = sphericalAngles[r0Lframe][[2]];
(*solPieceR10=SqA (β1l
      EllipticPi[(x2-x3)/(α1l+x2),JacobiAmplitude[ϒ0,β^2],β^2]/(α1l+x2)+β2l
      EllipticPi[(x2-x3)/(α2l+x2),JacobiAmplitude[ϒ0,β^2],β^2]/(α2l+x2));*)
solPieceR20 = Φ veϕ0 + ϵ ^ 2 (fϕ Sin[2 veϕ0] + gϕ Sin[3 veϕ0])
(*+ϵ (hϕ1 Sin[veϕ0-2 ϕs]+hϕ2 Sin[2 veϕ0-2 ϕs])*);

Torb = 2 π / nunscaled;


Return[N[{c, M, μ, ν, rvec, r, pvec, p, lvec, l, δ, h, ar, er, et, eϕ, n,
   nunscaled, ft, gt, fϕ, gϕ, Φ, EulMat, InvEulMat, drdt0, u0test, Absu0,
   u0, βeϕ, βer, veϕ0, ver0, t0, r0Lframe, rϕ0, solPieceR20, Torb}]];];


(*====================*)

Hflow2SListFixedSpins[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, AVG_, ORB_, tList_] :=
  Module[{pi, c, M, μ, ν, rvec, r, pvec, p, SS10, SS20, s1vec, s2vec, s1, s2, lvec, l,
    jvec, j, σ1, σ2, λ, δ1, δ2, δ, Δ, lseff, mathcalS2, h, Σ1, Σ2, s1vecLframe0,
    s2vecLframe0, δl20, ltilde2, δl2avg, lavg, a0, e0, spVecX, spVecY, ϕs, sp2, ar,
    er, et, eϕ, n, nunscaled, ft, gt, fϕ, gϕ, Φ, hr, hϕ1, hϕ2, d, A3, A2, A1, A0, A,
    x2, x3, x1, β, SqA, α1l, α2l, β1l, β2l, β3l, α1s1, α2s1, β1s1, β2s1, β3s1, EulMat,
    InvEulMat, drdt0, u0test, Absu0, u0, βeϕ, βer, veϕ0, ver0, t0, cosϰ10, Int30,
    sign, α, ϒ0, lvecJframe0, lξ10, solPieceL10, solPieceL20, s1vecJframe0, s1ξ10,
    solPieceS110, solPieceS120, JtoLframeEulMat0, r0Lframe, rϕ0, solPieceR10,
    solPieceR20, Nprec, Uprec, Tprec, Torb, freqcosϰ1, freqξ1L, computeForTime},

   pi = 3.141592653589793`;
```

```
{c, M, μ, ν, rvec, r, pvec, p, lvec, l, δ, h, ar, er, et, eϕ, n,
  nunscaled, ft, gt, fϕ, gϕ, Φ, EulMat, InvEulMat, drdt0, u0test, Absu0,
  u0, βeϕ, βer, veϕ0, ver0, t0, r0Lframe, rϕ0, solPieceR20, Torb} =
 computeBaseQuantitiesFixedSpin[m1, m2, R0, P0, S10, S20, ϵ, AVG, ORB];



(*Evolution for time t*)
computeForTime[tmax_] :=
 Module[{o, fveϕ, findu, umax, cosu, sinu, veϕ, ver, Int3, ϒ, JacAmp, cosϰ1,
   solPieceL1, lξ1sol, cosϰ2,
   lvecAzimuthAngle, lvecPolarAngle, LsolJframe, Lsol,
   solPieceS11, s1ξ1sol, cosγ,
   s1vecAzimuthAngle, s1vecPolarAngle, S1sol, S2sol,
   JtoLframeEulMat, s1vecLframe, s2vecLframe, solPieceR1, solPieceR2,
   solPieceR3, solPieceR4, rvecAzimuthAngleLframe, rMagnitude, Rsol},

  If[ORB == "Yes", o = 1];
  If[ORB == "No", o = 0];

  βeϕ = eϕ / (1 + Sqrt[1 - eϕ^2]);
  fveϕ = u + 2. ArcTan[(βeϕ Sin[u]) / (1 - βeϕ Cos[u])];

  findu[t_] := Module[{target, initialGuess}, target = nunscaled (t - t0);
    initialGuess = target;
    FindRoot[target == u - et Sin[u] + ϵ^2 (ft Sin[fveϕ] + gt (fveϕ - u)), {u,
      initialGuess}, Method → "Newton", WorkingPrecision → MachinePrecision]];

  umax = u /. findu[tmax];
  cosu = Cos[umax]; sinu = Sin[umax];
  veϕ = umax + 2. ArcTan[(βeϕ sinu) / (1 - βeϕ cosu)];
  ver = umax + 2. ArcTan[(βer sinu) / (1 - βer cosu)];

  (* ================================================================*)
  (*OVERRIDE:FREEZE ANGULAR MOMENTA TO INITIAL VALUES*)
  (* ================================================================*)
  Lsol = lvec * μ * M;
  S1sol = {0., 0., 0.} ;     (* s1vec*μ*M;*)
  S2sol = {0., 0., 0.} ;     (* s2vec*μ*M;*)
  LsolJframe = lvecJframe0 * μ * M;


  JtoLframeEulMat = IdentityMatrix[3];
```

```
      (*Kepler Equation Evaluation*)
      βeϕ = eϕ / (1 + Sqrt[1 - eϕ^2]);
      fveϕ = u + 2. ArcTan[(βeϕ Sin[u]) / (1 - βeϕ Cos[u])];
      findu[t_] := Module[{target, initialGuess},
        target = nunscaled (t - t0);
        initialGuess = target;
        FindRoot[target == u - et Sin[u] + ϵ^2 (ft Sin[fveϕ] + gt (fveϕ - u)), {u,
          initialGuess}, Method → "Newton", WorkingPrecision → MachinePrecision]];

umax = u /. findu[tmax];
sinu = Sin[umax]; cosu = Cos[umax];
veϕ = umax + 2. ArcTan[(βeϕ sinu) / (1 - βeϕ cosu)];

(*Recalculate purely orbital Azimuthal offset at t=
       0 to anchor the initial position cleanly*)
      solPieceR20Fixed = Φ veϕ0 + ϵ^2 (fϕ Sin[2 veϕ0] + gϕ Sin[3 veϕ0]);
solPieceR2 = Φ veϕ + ϵ^2 (fϕ Sin[2 veϕ] + gϕ Sin[3 veϕ]);

(*Azimuthal angle skips the spin-
       precession pieces (solPieceR1) and strictly uses in-plane dynamics*)
      rvecAzimuthAngleLframe = solPieceR2 + (rϕ0 - solPieceR20Fixed);

(*Radial magnitude strictly without Spin-Spin oscillations*)
      rMagnitude = ar (1 - er cosu);

Rsol = M (InvEulMat.Transpose[JtoLframeEulMat].
          vectorComponents[rMagnitude, Pi / 2, rvecAzimuthAngleLframe]);



      {(*1*)Rsol, (*2*)Lsol, (*3*)
       S1sol, (*4*)S2sol, (*5*)(*Tprec*)0, (*6*)Torb, (*7*)
       rvecAzimuthAngleLframe, (*8*)Transpose[JtoLframeEulMat], (*9*)
       vectorComponents[rMagnitude, Pi / 2, rvecAzimuthAngleLframe]}];

   Map[computeForTime, tList]];


(*========================*)

(*==*)(*EXECUTION WRAPPERS*)(*==*)
DataAn2S[AVG_, ORB_, t_, data_] := Block[{rvecAzimuthAngleLframe},
  Module[{m1, m2, R0, P0, S10, S20, ϵ}, {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   Hflow2SList[m1, m2, R0, P0, S10, S20, ϵ, AVG, ORB, t]]]
```

```
An2S[AVG_, ORB_, t_] := Association[# :> DataAn2S[AVG, ORB, t, #] & /@ Keys[Inputs]]


(*==*)(*EXECUTION WRAPPERS (2S FIXED SPINS)*)(*==*)
DataAn2SFixed[AVG_, ORB_, t_, data_] := Block[{rvecAzimuthAngleLframe},
  Module[{m1, m2, R0, P0, S10, S20, ϵ}, {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   Hflow2SListFixedSpins[m1, m2, R0, P0, S10, S20, ϵ, AVG, ORB, t]]]

An2SFixed[AVG_, ORB_, t_] :=
 Association[# :> DataAn2SFixed[AVG, ORB, t, #] & /@ Keys[Inputs]]
```

3. Inputs

A. Check Inputs

```
In[∘]:= CheckInputs[G_, m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, data_] :=
 Module[{M, μ, ν, rvec, r, pvec, p, SS10, SS20, s1vec, s2vec, s1, s2,
   s1ds2, lvec, l, jvec, j, Σ2, σ1, σ2, λ, δ1, δ2, dotlseff, seffdl, hN,
   h1, hso, h2, hss, H1, H2, HSS, varN, varSO, varSS, nN, aN, eN, ΦN, nSO,
   arSO, erSO, ΦSO, nSS, arSS, erSS, ΦSS, δhSO, δarSO, δerSO, δnSO, δΦSO,
   δhSS, δarSS, δerSS, δnSS, δΦSS, PNa, PNp, resultN, resultSO, resultSS,
   TprecN, TorbN, TprecSO, TorbSO, TprecSS, TorbSS, cosκ1, cosκ2, cosγ},

  M = m1 + m2; μ = m1 m2 / M; ν = μ / M;

  SS10 = Sqrt[ϵ] S10; SS20 = Sqrt[ϵ] S20;

  rvec = R0 / M; r = Norm[rvec]; pvec = P0 / μ; p = Norm[pvec];

  s1vec = SS10 / (μ M);
  s2vec = SS20 / (μ M);
  s1 = Norm[s1vec];
  s2 = Norm[s2vec];
  s1ds2 = s1vec.s2vec;

  lvec = Cross[rvec, pvec];
  l = Norm[lvec];
  jvec = lvec + s1vec + s2vec;
```

```
j = Norm[jvec];
δ1 = 2 ν (1 + 3 / 4 m2 / m1);
δ2 = 2 ν (1 + 3 / 4 m1 / m2);
dotlseff = lvec.(δ1 s1vec + δ2 s2vec);

(*Hamiltonian components*)
hN = p^2 / 2 - 1 / r;
h1 =
 ϵ (1 / (2 r^2) + 1 / 8 p^4 (-1 + 3 ν) - (Dot[rvec / r, pvec]^2 ν + p^2 (3 + ν)) / (2 r));
hso = + (dotlseff ϵ) / r^3;
h2 = ϵ^2 (- ((1 + 3 ν) / (4 r^3)) + 1 / 16 p^6 (1 - 5 ν + 5 ν^2) +
    (3 Dot[rvec / r, pvec]^2 ν + p^2 (5 + 8 ν)) / (2 r^2) + (-3 Dot[rvec / r, pvec]^4
        ν^2 - 2 Dot[rvec / r, pvec]^2 p^2 ν^2 + p^4 (5 - 20 ν - 3 ν^2)) / (8 r));
hss = - ((Dot[s1vec, s2vec] - 3 Dot[rvec / r, s1vec] × Dot[rvec / r, s2vec]) ϵ ν) / r^3 +
  (m2 (3 Dot[rvec / r, s1vec]^2 - s1^2) ϵ ν) / (2 m1 r^3) +
  (m1 (3 Dot[rvec / r, s2vec]^2 - s2^2) ϵ ν) / (2 m2 r^3);
seffdl = dotlseff + r^3 hss / ϵ;

H1 = hN + h1 + hso;
H2 = hN + h1 + hso + h2;
HSS = hN + h1 + hso + h2 + hss;

(*Compute QKP at different orders*)
varN = computeBaseQuantities1[m1, m2, R0, P0, S10, S20, ϵ, "N", "Yes"];
varSO = computeBaseQuantities1[m1, m2, R0, P0, S10, S20, ϵ, "SO", "Yes"];
varSS = computeBaseQuantities2S[m1, m2, R0, P0, S10, S20, ϵ, "No", "Yes"];

(*Extract values-Newtonian*)
nN = varN[[38]]; aN = varN[[42]]; eN = varN[[44]]; ΦN = 1;

(*Extract values-Spin-Orbit*)
nSO = varSO[[38]]; arSO = varSO[[42]]; erSO = varSO[[44]]; ΦSO = varSO[[54]];

(*Extract values-Spin-Spin*)
nSS = varSS[[45]]; arSS = varSS[[41]]; erSS = varSS[[42]]; ΦSS = varSS[[51]];

(*Compute differences N vs SO*)
δhSO = Abs[(H1 - hN) / H1];
δarSO = Abs[(arSO - aN) / arSO];
δerSO = Abs[(erSO - eN) / erSO];
δnSO = Abs[(nSO - nN) / nSO];
δΦSO = Abs[(ΦSO - ΦN) / ΦSO];

(*Compute differences SO vs SS*)
```

```
δhSS = Abs[(HSS - H1) / HSS];
δarSS = Abs[(arSS - arSO) / arSS];
δerSS = Abs[(erSS - erSO) / erSS];
δnSS = Abs[(nSS - nSO) / nSS];
δΦSS = Abs[(ΦSS - ΦSO) / ΦSS];

(*PN parameters*)
PNa = ϵ/r;
PNp = ϵ/(aN (1 - eN));
(*Angular variables*)
cosκ1 = s1vec.lvec / (s1 l);
cosκ2 = lvec.s2vec / (l s2);
cosγ = s1vec.s2vec / (s1 s2);

(*Get Tprec and Torb from Hflow functions*)
resultN = Hflow1List[m1, m2, R0, P0, S10, S20, ϵ, "N", "Yes", {0}][[1]];
TprecN = "Norb = " Tprec[data] / Torb[data];
TorbN = resultN[[6]];

resultSO = Hflow1List[m1, m2, R0, P0, S10, S20, ϵ, "SO", "Yes", {0}][[1]];
TprecSO = resultSO[[5]];
TorbSO = resultSO[[6]];

resultSS = Hflow2SList[m1, m2, R0, P0, S10, S20, ϵ, "No", "Yes", {0}][[1]];
TprecSS = resultSS[[5]];
TorbSS = resultSS[[6]];
(*Print organized table*)
Print@Grid[{{Style["Summary Table for Dataset: " data, Bold, 14],
    SpanFromLeft, SpanFromLeft, SpanFromLeft, SpanFromLeft, SpanFromLeft},
   (* ==================================================*)
   {Style["Initial configuration", Bold, 12], SpanFromLeft,
    SpanFromLeft, SpanFromLeft, SpanFromLeft, SpanFromLeft},
   {"q = m2/m1", "κ1 [°]", "κ2 [°]", "γ [°]", "PNa [%]", "PNp [%]"},
   {Round[m2 / m1, 0.01], Round[ArcCos[cosκ1] 180 / π, 0.1],
    Round[ArcCos[cosκ2] 180 / π, 0.1], Round[ArcCos[cosγ] 180 / π, 0.1],
    Round[100 PNa, 0.01], Round[100 PNp, 0.01]},
   (* ==================================================*)
   {Style["Spin strength", Bold, 12], SpanFromLeft,
    SpanFromLeft, SpanFromLeft, SpanFromLeft, SpanFromLeft},
   {"χ1", "χ2", "S1/(L ϵ^(1/2))", "S2/(L ϵ^(1/2))", "l·seff", SpanFromLeft},
   {Round[Norm[S10] / (G m1^2), 0.01],
    Round[Norm[S20] / (G m2^2), 0.01], Round[s1 / l / Sqrt[ϵ], 0.01],
```

```
      Round[s2 / l / Sqrt[ϵ], 0.01], Round[seffdl / Sqrt[ϵ], 0.001], SpanFromLeft},
     {Style["QKP Parameters", Bold, 12], SpanFromLeft,
      SpanFromLeft, SpanFromLeft, SpanFromLeft, SpanFromLeft},
     {"h", "n", "er", "Torb", "Tprec(Torb)"},
     {H2, nSS, erSS, TorbSS, TprecSS / TorbSS}},
    (* ==================================================*)
    Frame → All, Dividers → {{2 → Thick}, {1 → Thick, 5 → Thick, 9 → Thick}},
    Background →
     {None, {1 → LightBlue, 2 → LightYellow, 5 → LightYellow, 8 → LightYellow}},
    Alignment → {Center, Center},
    Spacings → {2, 1}, ItemSize → {{12, 12, 12, 12, 14, 14}, Automatic}]]
```

B. Numerical Inputs Random

1) Define Periods

```
In[ ]:= G = 1;(*Keep 1 because it is not put well everywhere*)

TprecData[data_] :=
  Module[{m1, m2, R0, P0, S10, S20, ϵ}, {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   Hflow2SList[m1, m2, R0, P0, S10, S20, ϵ, "No", "Yes", {0}][[1, 5]]];

Tprec := AssociationMap[TprecData, Keys[Inputs]] // Quiet;
(*One Precession Period*)

TorbData[data_] :=
  Module[{m1, m2, R0, P0, S10, S20, ϵ}, {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   Hflow2SList[m1, m2, R0, P0, S10, S20, ϵ, "No", "No", {0}][[1, 6]]];

Torb := AssociationMap[TorbData, Keys[Inputs]];(*One Orbital Period*)

PNparamData[data_] := Module[{m1, m2, R0, P0, S10, S20, ϵ, varN, aN, eN},
   {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
   varN = computeBaseQuantities1[m1, m2, R0, P0, S10, S20, ϵ, "N", "Yes"];
   aN = varN[[42]]; eN = varN[[44]];
   ϵ / (aN (1 - eN))];


PNparam := AssociationMap[PNparamData, Keys[Inputs]];(*One Orbital Period*)
```

2) Choose Data

In[ ]:=
```
CheckPhysicsConstraints[] := If[userData[[1]] ≤ userData[[2]],
   Print[Style["CRITICAL PHYSICS ERROR: Mass m1 must be strictly greater
       than m2. Evaluation halted.", "Message", Bold, Red]];
   Beep[];
   (*This tells the Front End to immediately drop all pending cells
    in the queue*)FrontEndExecute[FrontEndToken["EvaluatorAbort"]];];

(*Updated Outer Routine for the Summary Table*)
DisplaySummaryTable[] := (CheckPhysicsConstraints[];
  Inputs = <|data1 → userData|>;
  G = 1;
  Quiet[AssociationMap[Function[{data}, Module[
      {m1, m2, R0, P0, S10, S20, e}, {m1, m2, R0, P0, S10, S20, e} = Inputs[data];
      CheckInputs[1, m1, m2, R0, P0, S10, S20, e, data]]], Keys[Inputs]]];
  (* <--Semicolon added here*))
```

4. Summary Table

In[ ]:=
```
DisplaySummaryTable[]
```

Summary Table for Dataset:			
Initial configuration			
q = m2/m1	κ1 [°]	κ2 [°]	γ [°]
0.66	107.8	84.7	25.3
Spin strength			
χ1	χ2	$S_1/(L\ \epsilon^{1/2})$	$S_2/(L\ \epsilon^{1}$
0.34	0.48	0.43	0.27
QKP Parameters			
h	n	er	Torb
-0.311054	0.486138	0.381322	74.963

5. Numerical Solution

1. Code

In[ ]:=
```
(************************H = HN + H1PN + HSO************************)
ComputeNum1[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, ORB_, end_] :=
 Module[{o, G, μ, initialConditions, eqn1, eqn2, eqn3,
```

```
 eqn4, eqn5, sol1, equations, SS10, SS20, opts, t, EulMat},
If[ORB == "Yes", o = 1];
If[ORB == "No", o = 0];

G = 1;
μ = m1 m2 / (m1 + m2);
SS10 = S10 √ϵ ; SS20 = S20 √ϵ ;
EulMat = computeEulerMatrix[Cross[R0, P0] + SS10 + SS20];
(*initialConditions={R[0]==EulMat.R0,P[0]==EulMat.P0,
   S1[0]==EulMat.SS10,S2[0]==EulMat.SS20,L[0]==EulMat.Cross[R0,P0]};*)
initialConditions =
 {R[0] == R0, P[0] == P0, S1[0] == SS10, S2[0] == SS20, L[0] == Cross[R0, P0]};

eqn1 = R'[t] == + (P[t] / m1 + P[t] / m2) -
   o ϵ ( P[t] (P[t] . P[t]) / (2 m1 ^ 3) + P[t] (P[t] . P[t]) / (2 m2 ^ 3) +
      G R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ (3 / 2) + 7 G P[t] / (R[t] . R[t]) ^ (1 / 2) +
      3 G m1 P[t] / (m2 (R[t] . R[t]) ^ (1 / 2)) + 3 G m2 P[t] / (m1 (R[t] . R[t]) ^ (1 / 2))) -
   G ϵ (m2 (4 m1 + 3 m2) R[t] × S1[t] + m1 (3 m1 + 4 m2) R[t] × S2[t]) / (2 m1 m2 (R[t].R[t])^(3/2));
eqn2 = P'[t] == -(G m1 m2 R[t]) / (R[t].R[t])^(3/2) + o / (2 m1 m2 (R[t].R[t])^(5/2)) G ϵ (2 m1 m2 P[t].R[t] × R[t].
      R[t] × P[t] - 3 m1 m2 (P[t].R[t])^2 R[t] + 2 G m1^2 m2^2 (m1 + m2) √(R[t].R[t]) R[t] -
      (3 m1^2 + 7 m1 m2 + 3 m2^2) P[t].P[t] × R[t].R[t] × R[t]) - 1 / (8 m1^3 m2^3 (R[t].R[t])^(7/2))
    G ϵ (4 m1^2 m2^2 (m2 (4 m1 + 3 m2) P[t] × S1[t] + m1 (3 m1 + 4 m2) P[t] × S2[t])
      (R[t].R[t])^2 + 12 m1^2 m2^3 (4 m1 + 3 m2) P[t].R[t] × S1[t] × R[t].R[t] × R[t] +
      12 m1^3 m2^2 (3 m1 + 4 m2) P[t].R[t] × S2[t] × R[t].R[t] × R[t]);

eqn3 = S1'[t] == + (G (4 m1 + 3 m2) ϵ (R[t].S1[t] × P[t] - P[t].S1[t] × R[t])) / (2 m1 (R[t].R[t])^(3/2));

eqn4 = S2'[t] == + (G (4 m2 + 3 m1) ϵ (R[t].S2[t] × P[t] - P[t].S2[t] × R[t])) / (2 m2 (R[t].R[t])^(3/2));
eqn5 = L[t] == Cross[R[t], P[t]];
equations = {eqn1, eqn2, eqn3, eqn4, eqn5} ~ Join ~ initialConditions;
sol1 = NDSolve[equations, {R, P, S1, S2, L}, {t, 0, end},
   (*WorkingPrecision→workingprecision,*)AccuracyGoal → accuracyGoal,
   PrecisionGoal → precisionGoal, MaxSteps → maxSteps];
sol1]
(*H = HN + H1PN + HSO + H2PN *)
ComputeNum2[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, ORB_, end_] :=
 Module[{o, G, μ, initialConditions, eqn1, eqn2, eqn3,
```

```
 eqn4, eqn5, sol1, equations, SS10, SS20, opts, t, EulMat},

If[ORB == "Yes", o = 1];
If[ORB == "No", o = 0];

G = 1;
μ = m1 m2 / (m1 + m2);
SS10 = S10 √ϵ ; SS20 = S20 √ϵ ;
EulMat = computeEulerMatrix[Cross[R0, P0] + SS10 + SS20];
(*initialConditions={R[0]==EulMat.R0,P[0]==EulMat.P0,
   S1[0]==EulMat.SS10,S2[0]==EulMat.SS20,L[0]==EulMat.Cross[R0,P0]};*)
initialConditions = {R[0] == SetPrecision[R0, 20],
  P[0] == SetPrecision[P0, 20], S1[0] == SetPrecision[SS10, 20],
  S2[0] == SetPrecision[SS20, 20], L[0] == SetPrecision[Cross[R0, P0], 20]};

eqn1 = R'[t] == + (P[t] / m1 + P[t] / m2) -
   o ϵ ( P[t] (P[t] . P[t]) / (2 m1 ^ 3) + P[t] (P[t] . P[t]) / (2 m2 ^ 3) +
      G R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ (3 / 2) + 7 G P[t] / (R[t] . R[t]) ^ (1 / 2) +
      3 G m1 P[t] / (m2 (R[t] . R[t]) ^ (1 / 2)) + 3 G m2 P[t] / (m1 (R[t] . R[t]) ^ (1 / 2))) -
   G ϵ (m2 (4 m1 + 3 m2) R[t] × S1[t] + m1 (3 m1 + 4 m2) R[t] × S2[t])
   ------------------------------------------------------------------ +
                   2 m1 m2 (R[t].R[t])^(3/2)
   o ϵ ^2 (3 P[t] (P[t] . P[t]) ^2 / (8 m1 ^ 5) + (3 P[t] (P[t] . P[t]) ^2) / (8 m2 ^ 5) -
      3 G R[t] (P[t] . R[t]) ^3 / (2 m1 m2 (R[t] . R[t]) ^ (5 / 2)) +
      3 (G ^ 2) m1 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 +
      3 (G ^ 2) m2 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 -
      G R[t] (P[t] . P[t]) (P[t] . R[t]) / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) -
      G P[t] (P[t] . R[t]) ^ 2 / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) +
      23 G ^ 2 m1 P[t] / (R[t] . R[t]) + 5 G ^ 2 m1 ^ 2 P[t] / (m2 (R[t] . R[t])) +
      23 G ^ 2 m2 P[t] / (R[t] . R[t]) + 5 G ^ 2 m2 ^ 2 P[t] / (m1 (R[t] . R[t])) +
      5 G m1 P[t] (P[t] . P[t]) / (2 m2 ^ 3 (R[t] . R[t]) ^ (1 / 2)) +
      5 G m2 P[t] (P[t] . P[t]) / (2 m1 ^ 3 (R[t] . R[t]) ^ (1 / 2)) -
      13 G P[t] (P[t] . P[t]) / (2 m2 m1 (R[t] . R[t]) ^ (1 / 2)));

                   G m1 m2 R[t]
eqn2 = P'[t] == - ------------- +
                 (R[t].R[t])^(3/2)
   o (G ϵ (2 m1 m2 P[t].R[t] × R[t].R[t] × P[t] - 3 m1 m2 (P[t].R[t])^2 R[t] +
         2 G m1^2 m2^2 (m1 + m2) √(R[t].R[t]) R[t] - (3 m1^2 + 7 m1 m2 + 3 m2^2) P[t].P[t] ×
          R[t].R[t] × R[t])) / (2 m1 m2 (R[t].R[t])^(5/2)) - 1/(8 m1^3 m2^3 (R[t].R[t])^(7/2)) G ϵ
     (4 m1^2 m2^2 (m2 (4 m1 + 3 m2) P[t] × S1[t] + m1 (3 m1 + 4 m2) P[t] × S2[t]) (R[t].R[t])^2 +
       12 m1^2 m2^3 (4 m1 + 3 m2) P[t].R[t] × S1[t] × R[t].R[t] × R[t] +
       12 m1^3 m2^2 (3 m1 + 4 m2) P[t].R[t] × S2[t] × R[t].R[t] × R[t]) -
```

```
        o
  ----------------------------------- G ϵ^2 (-12 m1^2 m2^2 (P[t].R[t])^3 R[t].R[t] × P[t] +
  8 m1^3 m2^3 (R[t].R[t])^(7/2)
      24 G m1^3 m2^3 (m1 + m2) P[t].R[t] (R[t].R[t])^(3/2) P[t] -
      4 m1^2 m2^2 P[t].P[t] × P[t].R[t] (R[t].R[t])^2 P[t] + 15 m1^2 m2^2 (P[t].R[t])^4 R[t] -
      48 G m1^3 m2^3 (m1 + m2) (P[t].R[t])^2 Sqrt[R[t].R[t]] R[t] +
      6 G^2 m1^6 m2^4 R[t].R[t] × R[t] + 30 G^2 m1^5 m2^5 R[t].R[t] × R[t] +
      6 G^2 m1^4 m2^6 R[t].R[t] × R[t] + 6 m1^2 m2^2 P[t].P[t] (P[t].R[t])^2 R[t].R[t] × R[t] -
      8 G m1^2 m2^2 (m1 + m2) (8 m1 m2 + 5 (m1 + m2)^2) P[t].P[t] (R[t].R[t])^(3/2) R[t] +
      (-5 m1^4 + 13 m1^2 m2^2 - 5 m2^4) (P[t].P[t])^2 (R[t].R[t])^2 R[t]);

  eqn3 = S1'[t] == + G (4 m1 + 3 m2) ϵ (R[t].S1[t] × P[t] - P[t].S1[t] × R[t]) / (2 m1 (R[t].R[t])^(3/2));


  eqn4 = S2'[t] == + G (4 m2 + 3 m1) ϵ (R[t].S2[t] × P[t] - P[t].S2[t] × R[t]) / (2 m2 (R[t].R[t])^(3/2));

  eqn5 = L[t] == Cross[R[t], P[t]];
  equations = {eqn1, eqn2, eqn3, eqn4, eqn5} ~ Join ~ initialConditions;
  sol1 = NDSolve[equations, {R, P, S1, S2, L}, {t, 0, end},
    (*WorkingPrecision→workingprecision,*) AccuracyGoal → accuracyGoal,
    PrecisionGoal → precisionGoal, MaxSteps → maxSteps];
  sol1
]

(*H = HN + H1PN + H2PN *)
ComputeNumOrb[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, ORB_, end_] :=
 Module[{o, G, μ, initialConditions, eqn1, eqn2, eqn3,
   eqn4, eqn5, sol1, equations, SS10, SS20, opts, t, EulMat},

  If[ORB == "Yes", o = 1];
  If[ORB == "No", o = 0];

  G = 1;
  μ = m1 m2 / (m1 + m2);
  SS10 = S10 Sqrt[ϵ]; SS20 = S20 Sqrt[ϵ];
  EulMat = computeEulerMatrix[Cross[R0, P0] + SS10 + SS20];
  initialConditions = {R[0] == SetPrecision[R0, 20],
    P[0] == SetPrecision[P0, 20], S1[0] == SetPrecision[SS10, 20],
    S2[0] == SetPrecision[SS20, 20], L[0] == SetPrecision[Cross[R0, P0], 20]};

  eqn1 = R'[t] == + (P[t] / m1 + P[t] / m2) -
```

```
    o ϵ ( P[t] (P[t] . P[t]) / (2 m1 ^ 3) + P[t] (P[t] . P[t]) / (2 m2 ^ 3) +
       G R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ (3 / 2) + 7 G P[t] / (R[t] . R[t]) ^ (1 / 2) + 3 G
        m1 P[t] / (m2 (R[t] . R[t]) ^ (1 / 2)) + 3 G m2 P[t] / (m1 (R[t] . R[t]) ^ (1 / 2))) +
    o ϵ ^ 2 (3 P[t] (P[t] . P[t]) ^ 2 / (8 m1 ^ 5) + (3 P[t] (P[t] . P[t]) ^ 2) / (8 m2 ^ 5) -
       3 G R[t] (P[t] . R[t]) ^ 3 / (2 m1 m2 (R[t] . R[t]) ^ (5 / 2)) +
       3 (G ^ 2) m1 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 +
       3 (G ^ 2) m2 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 -
       G R[t] (P[t] . P[t]) (P[t] . R[t]) / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) -
       G P[t] (P[t] . R[t]) ^ 2 / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) +
       23 G ^ 2 m1 P[t] / (R[t] . R[t]) + 5 G ^ 2 m1 ^ 2 P[t] / (m2 (R[t] . R[t])) +
       23 G ^ 2 m2 P[t] / (R[t] . R[t]) + 5 G ^ 2 m2 ^ 2 P[t] / (m1 (R[t] . R[t])) +
       5 G m1 P[t] (P[t] . P[t]) / (2 m2 ^ 3 (R[t] . R[t]) ^ (1 / 2)) +
       5 G m2 P[t] (P[t] . P[t]) / (2 m1 ^ 3 (R[t] . R[t]) ^ (1 / 2)) -
       13 G P[t] (P[t] . P[t]) / (2 m2 m1 (R[t] . R[t]) ^ (1 / 2)));
```

$$
\begin{aligned}
\text{eqn2} = P'[t] == & -\frac{G\, m1\, m2\, R[t]}{(R[t].R[t])^{3/2}} + \\
& o\Big(G\,\epsilon\Big(2\, m1\, m2\, P[t].R[t] \times R[t].R[t] \times P[t] - 3\, m1\, m2\, (P[t].R[t])^2\, R[t] + \\
& \quad 2\, G\, m1^2\, m2^2\, (m1 + m2)\, \sqrt{R[t].R[t]}\, R[t] - \left(3\, m1^2 + 7\, m1\, m2 + 3\, m2^2\right) \\
& \quad P[t].P[t] \times R[t].R[t] \times R[t]\Big)\Big) \Big/ \left(2\, m1\, m2\, (R[t].R[t])^{5/2}\right) - \\
& \frac{o}{8\, m1^3\, m2^3\, (R[t].R[t])^{7/2}}\, G\, \epsilon^2 \Big(-12\, m1^2\, m2^2\, (P[t].R[t])^3\, R[t].R[t] \times P[t] + \\
& \quad 24\, G\, m1^3\, m2^3\, (m1 + m2)\, P[t].R[t]\, (R[t].R[t])^{3/2}\, P[t] - \\
& \quad 4\, m1^2\, m2^2\, P[t].P[t] \times P[t].R[t]\, (R[t].R[t])^2\, P[t] + 15\, m1^2\, m2^2\, (P[t].R[t])^4\, R[t] - \\
& \quad 48\, G\, m1^3\, m2^3\, (m1 + m2)\, (P[t].R[t])^2\, \sqrt{R[t].R[t]}\, R[t] + \\
& \quad 6\, G^2\, m1^6\, m2^4\, R[t].R[t] \times R[t] + 30\, G^2\, m1^5\, m2^5\, R[t].R[t] \times R[t] + \\
& \quad 6\, G^2\, m1^4\, m2^6\, R[t].R[t] \times R[t] + 6\, m1^2\, m2^2\, P[t].P[t]\, (P[t].R[t])^2\, R[t].R[t] \times R[t] - \\
& \quad 8\, G\, m1^2\, m2^2\, (m1 + m2)\, \left(8\, m1\, m2 + 5\, (m1 + m2)^2\right) P[t].P[t]\, (R[t].R[t])^{3/2}\, R[t] + \\
& \quad \left(-5\, m1^4 + 13\, m1^2\, m2^2 - 5\, m2^4\right) (P[t].P[t])^2\, (R[t].R[t])^2\, R[t]\Big);
\end{aligned}
$$

```
eqn3 = S1'[t] == 0;


eqn4 = S2'[t] == 0;

eqn5 = L[t] == Cross[R[t], P[t]];
equations = {eqn1, eqn2, eqn3, eqn4, eqn5} ~ Join ~ initialConditions;
sol1 = NDSolve[equations, {R, P, S1, S2, L}, {t, 0, end},
  (*WorkingPrecision→workingprecision,*) AccuracyGoal → accuracyGoal,
  PrecisionGoal → precisionGoal, MaxSteps → maxSteps];
sol1
]
```

```
(*H = HN + H1PN + HSO + H2PN + H2PNSS*)
ComputeNum2S[m1_, m2_, R0_, P0_, S10_, S20_, ϵ_, ORB_, end_] :=
 Module[{o, G, μ, initialConditions, eqn1, eqn2, eqn3,
   eqn4, eqn5, sol1, equations, SS10, SS20, opts, t, EulMat},

  If[ORB == "Yes", o = 1];
  If[ORB == "No", o = 0];

  G = 1;
  μ = m1 m2 / (m1 + m2);
  SS10 = S10 √ϵ ; SS20 = S20 √ϵ ;
  EulMat = computeEulerMatrix[Cross[R0, P0] + SS10 + SS20];
  (*initialConditions={R[0]==EulMat.R0,P[0]==EulMat.P0,
     S1[0]==EulMat.SS10,S2[0]==EulMat.SS20,L[0]==EulMat.Cross[R0,P0]};*)
  initialConditions = {R[0] == SetPrecision[R0, 20],
    P[0] == SetPrecision[P0, 20], S1[0] == SetPrecision[SS10, 20],
    S2[0] == SetPrecision[SS20, 20], L[0] == SetPrecision[Cross[R0, P0], 20]};

  eqn1 = R'[t] == +(P[t] / m1 + P[t] / m2) -
     o ϵ ( P[t] (P[t] . P[t]) / (2 m1 ^ 3) + P[t] (P[t] . P[t]) / (2 m2 ^ 3) +
        G R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ (3 / 2) + 7 G P[t] / (R[t] . R[t]) ^ (1 / 2) +
        3 G m1 P[t] / (m2 (R[t] . R[t]) ^ (1 / 2)) + 3 G m2 P[t] / (m1 (R[t] . R[t]) ^ (1 / 2))) -
     G ϵ (m2 (4 m1 + 3 m2) R[t] × S1[t] + m1 (3 m1 + 4 m2) R[t] × S2[t])
     ----------------------------------------------------------------- +
                    2 m1 m2 (R[t].R[t])^(3/2)
     o ϵ ^2 (3 P[t] (P[t] . P[t]) ^2 / (8 m1 ^ 5) + (3 P[t] (P[t] . P[t]) ^2) / (8 m2 ^ 5) -
        3 G R[t] (P[t] . R[t]) ^3 / (2 m1 m2 (R[t] . R[t]) ^ (5 / 2)) +
        3 (G ^ 2) m1 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 +
        3 (G ^ 2) m2 R[t] (P[t] . R[t]) / (R[t] . R[t]) ^ 2 -
        G R[t] (P[t] . P[t]) (P[t] . R[t]) / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) -
        G P[t] (P[t] . R[t]) ^ 2 / (2 m1 m2 (R[t] . R[t]) ^ (3 / 2)) +
        23 G ^ 2 m1 P[t] / (R[t] . R[t]) + 5 G ^ 2 m1 ^ 2 P[t] / (m2 (R[t] . R[t])) +
        23 G ^ 2 m2 P[t] / (R[t] . R[t]) + 5 G ^ 2 m2 ^ 2 P[t] / (m1 (R[t] . R[t])) +
        5 G m1 P[t] (P[t] . P[t]) / (2 m2 ^ 3 (R[t] . R[t]) ^ (1 / 2)) +
        5 G m2 P[t] (P[t] . P[t]) / (2 m1 ^ 3 (R[t] . R[t]) ^ (1 / 2)) -
        13 G P[t] (P[t] . P[t]) / (2 m2 m1 (R[t] . R[t]) ^ (1 / 2)));

                                  G m1 m2 R[t]
eqn2 = P'[t] == -----------------------------------------------------------------
```

```
        R[t].R[t] × R[t])) / (2 m1 m2 (R[t].R[t])^(5/2)) - 1/(8 m1^3 m2^3 (R[t].R[t])^(7/2)) G ϵ
      (4 m1^2 m2^2 (m2 (4 m1 + 3 m2) P[t] × S1[t] + m1 (3 m1 + 4 m2) P[t] × S2[t]) (R[t].R[t])^2 +
        12 m1^2 m2^3 (4 m1 + 3 m2) P[t].R[t] × S1[t] × R[t].R[t] × R[t] +
        12 m1^3 m2^2 (3 m1 + 4 m2) P[t].R[t] × S2[t] × R[t].R[t] × R[t]) -
      o/(8 m1^3 m2^3 (R[t].R[t])^(7/2)) G ϵ^2 (-12 m1^2 m2^2 (P[t].R[t])^3 R[t].R[t] × P[t] +
        24 G m1^3 m2^3 (m1 + m2) P[t].R[t] (R[t].R[t])^(3/2) P[t] -
        4 m1^2 m2^2 P[t].P[t] × P[t].R[t] (R[t].R[t])^2 P[t] + 15 m1^2 m2^2 (P[t].R[t])^4 R[t] -
        48 G m1^3 m2^3 (m1 + m2) (P[t].R[t])^2 Sqrt[R[t].R[t]] R[t] +
        6 G^2 m1^6 m2^4 R[t].R[t] × R[t] + 30 G^2 m1^5 m2^5 R[t].R[t] × R[t] +
        6 G^2 m1^4 m2^6 R[t].R[t] × R[t] + 6 m1^2 m2^2 P[t].P[t] (P[t].R[t])^2 R[t].R[t] × R[t] -
        8 G m1^2 m2^2 (m1 + m2) (8 m1 m2 + 5 (m1 + m2)^2) P[t].P[t] (R[t].R[t])^(3/2) R[t] +
        (-5 m1^4 + 13 m1^2 m2^2 - 5 m2^4) (P[t].P[t])^2 (R[t].R[t])^2 R[t]) -
      1/(2 m1 m2 (R[t].R[t])^(7/2)) 3 G ϵ (-5 m2^2 (R[t].S1[t])^2 R[t] - 5 m1^2 (R[t].S2[t])^2 R[t] +
        R[t].R[t] (m2^2 S1[t].S1[t] + 2 m1 m2 S1[t].S2[t] + m1^2 S2[t].S2[t]) R[t] +
        2 m1 R[t].R[t] × R[t].S2[t] (m2 S1[t] + m1 S2[t]) +
        2 m2 R[t].S1[t] (-5 m1 R[t].S2[t] × R[t] + R[t].R[t] (m2 S1[t] + m1 S2[t])));

eqn3 =
  S1'[t] == + (G (4 m1 + 3 m2) ϵ (R[t].S1[t] × P[t] - P[t].S1[t] × R[t]))/(2 m1 (R[t].R[t])^(3/2)) + 1/(m1 (R[t].R[t])^(5/2))
     G ϵ (m1 S1[t] × S2[t] × R[t].R[t] + 3 R[t] × S1[t] (m2 R[t].S1[t] + m1 R[t].S2[t]));


  eqn4 = S2'[t] ==
    + (G (4 m2 + 3 m1) ϵ (R[t].S2[t] × P[t] - P[t].S2[t] × R[t]))/(2 m2 (R[t].R[t])^(3/2)) + 1/(m2 (R[t].R[t])^(5/2)) G ϵ
      (-m2 S1[t] × S2[t] × R[t].R[t] + 3 R[t] × S2[t] (m1 R[t].S2[t] + m2 R[t].S1[t]));

  eqn5 = L[t] == Cross[R[t], P[t]];
  equations = {eqn1, eqn2, eqn3, eqn4, eqn5} ~ Join ~ initialConditions;
  sol1 = NDSolve[equations, {R, P, S1, S2, L}, {t, 0, end},
    (*WorkingPrecision→workingprecision,*)AccuracyGoal → accuracyGoal,
    PrecisionGoal → precisionGoal, MaxSteps → maxSteps];
  sol1
]
```

2. Compute

```
In[∘]:= (************************H = HN + H1PN + HSO************************)
DataNum1[data_] := Module[{m1, m2, R0, P0, S10, S20, ϵ},
  {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
  ComputeNum1[m1, m2, R0, P0, S10, S20, ϵ, NumOrb, end[data]]]
Num1 = AssociationMap[DataNum1, Keys[Inputs]];

In[∘]:= (*(************************H = HN + H1PN + HSO + H2PN************************)
DataNum2[data_]:=Module[{m1,m2,R0,P0,S10,S20,ϵ},
     {m1,m2,R0,P0,S10,S20,ϵ}=Inputs[data];
     ComputeNum2[m1,m2,R0,P0,S10,S20,ϵ, NumOrb,end[data]]]
    Num2=AssociationMap[DataNum2,Keys[Inputs]];*)

In[∘]:= (************************H =
 HN + H1PN + HSO + H2PN + HSS************************)
DataNum2S[data_] := Module[{m1, m2, R0, P0, S10, S20, ϵ},
  {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
  ComputeNum2S[m1, m2, R0, P0, S10, S20, ϵ, NumOrb, end[data]]]

Num2S = AssociationMap[DataNum2S, Keys[Inputs]];


In[∘]:= DataNumOrb[data_] := Module[{m1, m2, R0, P0, S10, S20, ϵ},
  {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data];
  ComputeNumOrb[m1, m2, R0, P0, S10, S20, ϵ, NumOrb, end[data]]]
NumOrb = AssociationMap[DataNumOrb, Keys[Inputs]];
```

3. AzimuthalRvectorLframe

```
In[•]:= (*Function to compute the azimuthal angle of R vector in L frame*)
AzimuthalRvectorLframe[t_, NumModel_, data_] :=
 Module[{vectorComponents, rvecLabFrame, pvecLabFrame, s1vecLabFrame,
   s2vecLabFrame, lvecLabFrame, jvecLabFrame, LabtoJEulMat, θJ, ϕJ,
   lsolJframe, lx, ly, lz, ξ1L, ξ2L, JtoLEulMat, rvecLframe, angles, rNorm,
   rvecAzimuthAngleLframe, RvecLabFrame, m1, m2, R0, P0, S10, S20, ϵ, M, μ},

  {m1, m2, R0, P0, S10, S20, ϵ} = Inputs[data]; M = m1 + m2; μ = m1 m2 / M;

  (*Numerical Values at time t in Lab frame*)
  rvecLabFrame = (R[t] /. NumModel[data][[1]]) / (M);
  pvecLabFrame = (P[t] /. NumModel[data][[1]]) / (μ);
  s1vecLabFrame = (S1[t] /. NumModel[data][[1]]) / (μ M);
  s2vecLabFrame = (S2[t] /. NumModel[data][[1]]) / (μ M);
  lvecLabFrame = (L[t] /. NumModel[data][[1]]) / (μ M);
  jvecLabFrame = lvecLabFrame + s1vecLabFrame + s2vecLabFrame;
  rNorm = Norm@rvecLabFrame;

  (*Euler Matrix from Lab to J frame at all times*)
  {θJ, ϕJ} = sphericalAngles[jvecLabFrame] + {0, π / 2};
  LabtoJEulMat = {{Cos[ϕJ], Sin[ϕJ], 0}, {-Sin[ϕJ] Cos[θJ], Cos[ϕJ] Cos[θJ],
     Sin[θJ]}, {Sin[ϕJ] Sin[θJ], -Cos[ϕJ] Sin[θJ], Cos[θJ]}};
  (*L vector in J frame*)
  lsolJframe = LabtoJEulMat.lvecLabFrame;

  (*Euler Matrix from J to L frame at time t*)
  {lx, ly, lz} = lsolJframe;
  {ξ2L, ξ1L} = sphericalAngles[{lx, ly, lz}] + {0, π / 2};
  JtoLEulMat =
   { {Cos[ξ1L], Sin[ξ1L], 0}, {-Sin[ξ1L] Cos[ξ2L], Cos[ξ1L] Cos[ξ2L], Sin[ξ2L]},
    {Sin[ξ1L] Sin[ξ2L], -Cos[ξ1L] Sin[ξ2L], Cos[ξ2L]}};
  (*r vector in L frame*)
  rvecLframe = JtoLEulMat.LabtoJEulMat.rvecLabFrame;

  (*AzimuthAngle of rvec in Lframe : what we want*)
  rvecAzimuthAngleLframe = sphericalAngles[rvecLframe][[2]];

  (*Conistency check, have to recover R0*)
  RvecLabFrame = M Transpose[LabtoJEulMat] . Inverse[JtoLEulMat].
     vectorComponents[{rNorm, π / 2, rvecAzimuthAngleLframe}];

  Return [{rvecAzimuthAngleLframe, Transpose@JtoLEulMat, rvecLframe}];]
```

6. Final Plots

A) Code

In[ ]:=
```
(*==*)(*COLORS& VECTOR NAMES*)(*==*)
CBlue = RGBColor[0.24, 0.6, 0.8];
COrange = RGBColor[0.95, 0.627, 0.1425];
CGreen = RGBColor[0.455, 0.7, 0.21];
CRed = RGBColor[0.982526, 0.385626, 0.209179];
CPurple = RGBColor[0.578, 0.51, 0.85];
CBrown = RGBColor[0.772079, 0.431554, 0.102387];

Vector = {R, L, S1, S2};
Legend = {{"Rx", "Ry", "Rz"}, {"Lx", "Ly", "Lz"},
   {"S1x", "S1y", "S1z"}, {"S2x", "S2y", "S2z"}};
LegendNorm = {"R", "L", "S1", "S2"};
LegendLtoJ = {{"(1,1)", "(1,2)", "(1,3)"},
   {"(2,1)", "(2,2)", "(2,3)"}, {"(3,1)", "(3,2)", "(3,3)"}};

Lhat[v_] := v / Norm[v];

(*==*)(*MODELS + Choose Orbital Terms*)(*==*)
PlotAn1[t_, data_] := An1[QKP, Orb, {t}][data]⟦1⟧⟦NumVec, NumComp⟧;
PlotAn2S[t_, data_] := An2S["No", Orb, {t}][data]⟦1⟧⟦NumVec, NumComp⟧;
PlotAn2Savg[t_, data_] := An2S["Yes", Orb, {t}][data]⟦1⟧⟦NumVec, NumComp⟧;
PlotNum1[t_, data_] := (Vector⟦NumVec⟧[t] /. Num1[data]⟦1⟧)⟦NumComp⟧;
PlotNum2S[t_, data_] := (Vector⟦NumVec⟧[t] /. Num2S[data]⟦1⟧)⟦NumComp⟧;
PlotNumOrb[t_, data_] := (Vector⟦NumVec⟧[t] /. NumOrb[data]⟦1⟧)⟦NumComp⟧;

(*ADD THIS LINE*)
PlotAn2SFixed[t_, data_] := An2SFixed["No", Orb, {t}][data]⟦1⟧⟦NumVec, NumComp⟧;

PlotAn1Norm[t_, data_] := Norm[An1[QKP, Orb, {t}][data]⟦1⟧⟦NumVec⟧];
PlotAn2SNorm[t_, data_] := Norm[An2S["No", Orb, {t}][data]⟦1⟧⟦NumVec⟧];
PlotNum1Norm[t_, data_] := Norm[(Vector⟦NumVec⟧[t] /. Num1[data]⟦1⟧)];
PlotNum2SNorm[t_, data_] := Norm[(Vector⟦NumVec⟧[t] /. Num2S[data]⟦1⟧)];
PlotNumOrbNorm[t_, data_] := Norm[(Vector⟦NumVec⟧[t] /. NumOrb[data]⟦1⟧)];

azNum1[t_, data_] := Mod[AzimuthalRvectorLframe[t, Num1, data]⟦1⟧, 2 π];
azNum2S[t_, data_] := Mod[AzimuthalRvectorLframe[t, Num2S, data]⟦1⟧, 2 π];
azAn1[t_, data_] := Mod[An1[QKP, Orb, {t}][data]⟦1, 7⟧, 2 π];
azAn2S[t_, data_] := Mod[An2S["No", Orb, {t}][data]⟦1, 7⟧, 2 π];
```

```
LtoJNum1[t_, data_] := AzimuthalRvectorLframe[t, Num1, data]⟦2⟧⟦Compi, Compj⟧;
LtoJNum2S[t_, data_] := AzimuthalRvectorLframe[t, Num2S, data]⟦2⟧⟦Compi, Compj⟧;
LtoJAn1[t_, data_] := An1[QKP, Orb, {t}][data]⟦1, 8⟧⟦Compi, Compj⟧;
LtoJAn2S[t_, data_] := An2S["No", Orb, {t}][data]⟦1, 8⟧⟦Compi, Compj⟧;

RLframeNum1[t_, data_] := AzimuthalRvectorLframe[t, Num1, data]⟦3⟧⟦NumComp⟧;
RLframeNum2S[t_, data_] := AzimuthalRvectorLframe[t, Num2S, data]⟦3⟧⟦NumComp⟧;
RLframeAn1[t_, data_] := An1[QKP, Orb, {t}][data]⟦1, 9⟧⟦NumComp⟧;
RLframeAn2S[t_, data_] := An2S["No", Orb, {t}][data]⟦1, 9⟧⟦NumComp⟧;

NumModel = Num2S;

(*PLOTTING FUNCTIONS*)
DefPlot[data_, vecNum_, compNum_, tmin_, tmax_,
   plotIndices_List, numPoints_ : 10] := Module[{allFunctions,
    functionLabels, selectedFunctions, TorbVal, tValues, tScaledData},
   allFunctions = {PlotAn1, PlotAn2S,
     PlotAn2Savg, PlotNum1, PlotNum2S, PlotAn2SFixed, PlotNumOrb};
functionLabels = {"1.5PN AN", "2PN AN", "2PN avg AN",
     "1.5PN NUM", "2PN NUM", "2PN FIXED", "2PN FIXED NUM"};
   selectedFunctions = allFunctions⟦plotIndices⟧;
   TorbVal = Torb[data];
   tValues = Table[tmin + i * (tmax - tmin) / (numPoints - 1), {i, 0, numPoints - 1}];
   tScaledData = Table[Table[{t / TorbVal, selectedFunctions⟦i⟧[t, data]},
      {t, tValues}], {i, Length[selectedFunctions]}];
   ListLinePlot[tScaledData, Frame → True, FrameLabel →
     {"t / \!\(\*SubscriptBox[\(T\), \(orb\)]\)", Legend⟦vecNum, compNum⟧},
    PlotRange → All, FrameStyle → Directive[Black, Thin],
    TicksStyle → Directive[FontSize → 12],
    PlotStyle → {{Continue, CBlue}, {Continue, COrange},
       {Continue, CGreen}, {Dashed, CPurple}, {Dashed, CRed},
       {Continue, CBrown}, {Dashed, Green}}⟦plotIndices⟧,
    PlotLegends → Placed[LineLegend[Style[#, FontSize → 16] & /@
        functionLabels⟦plotIndices⟧, LegendLayout → {"Row", 1}], Top]]];


DefPlotNorm[data_, vecNum_, compNum_, tmin_, tmax_,
   plotIndices_List, numPoints_ : 10] := Module[{allFunctions,
    functionLabels, selectedFunctions, TorbVal, tValues, tScaledData},
   allFunctions =
    {PlotAn1, PlotAn2S, PlotAn2Savg, PlotNum1, PlotNum2S, PlotAn2SFixed};
functionLabels =
    {"1.5PN AN", "2PN AN", "2PN avg AN", "1.5PN NUM", "2PN NUM", "2PN FIXED"};
```

```
    selectedFunctions = allFunctions⟦plotIndices⟧;
    TorbVal = Torb[data];
    tValues = Table[tmin + i * (tmax - tmin) / (numPoints - 1), {i, 0, numPoints - 1}];
    tScaledData = Table[Table[{t / TorbVal, selectedFunctions⟦i⟧[t, data]},
       {t, tValues}], {i, Length[selectedFunctions]}];
    ListLinePlot[tScaledData, Frame → True, FrameLabel →
      {"t / \!\(\*SubscriptBox[\(T\), \(orb\)]\)", LegendNorm⟦vecNum⟧},
     PlotRange → All, FrameStyle → Directive[Black, Thin],
     TicksStyle → Directive[FontSize → 12],
     PlotStyle → {{Continue, CBlue}, {Continue, COrange}, {Continue, CGreen},
        {Dashed, CPurple}, {Dashed, CRed}, {Dashed, CBrown}}⟦plotIndices⟧,
     PlotLegends → Placed[LineLegend[Style[#, FontSize → 16] & /@
         functionLabels⟦plotIndices⟧, LegendLayout → {"Row", 1}], Top]]];

DefPlotAz[data_, tmin_, tmax_, plotIndices_List, numPoints_ : 10] :=
  Module[{allFunctions, functionLabels,
     selectedFunctions, TorbVal, tValues, tScaledData},
    allFunctions =
     {PlotAn1, PlotAn2S, PlotAn2Savg, PlotNum1, PlotNum2S, PlotAn2SFixed};
functionLabels =
     {"1.5PN AN", "2PN AN", "2PN avg AN", "1.5PN NUM", "2PN NUM", "2PN FIXED"};
    selectedFunctions = allFunctions⟦plotIndices⟧;
    TorbVal = Torb[data];
    tValues = Table[tmin + i * (tmax - tmin) / (numPoints - 1), {i, 0, numPoints - 1}];
    tScaledData = Table[Table[{t / TorbVal, selectedFunctions⟦i⟧[t, data]},
       {t, tValues}], {i, Length[selectedFunctions]}];
    ListLinePlot[tScaledData, Frame → True,
     FrameLabel → {"t / \!\(\*SubscriptBox[\(T\), \(orb\)]\)", "ϕ"},
     PlotRange → All, FrameStyle → Directive[Black, Thin],
     TicksStyle → Directive[FontSize → 12],
     PlotStyle → {{Continue, CBlue}, {Continue, COrange}, {Continue, CGreen},
        {Dashed, CPurple}, {Dashed, CRed}, {Dashed, CBrown}}⟦plotIndices⟧,
     PlotLegends → Placed[LineLegend[Style[#, FontSize → 16] & /@
         functionLabels⟦plotIndices⟧, LegendLayout → {"Row", 1}], Top]]];

DefPlotLtoJ[data_, Compi_, Compj_, tmin_, tmax_,
    plotIndices_List, numPoints_ : 10] := Module[{allFunctions,
     functionLabels, selectedFunctions, TorbVal, tValues, tScaledData},
    allFunctions =
     {PlotAn1, PlotAn2S, PlotAn2Savg, PlotNum1, PlotNum2S, PlotAn2SFixed};
functionLabels =
     {"1.5PN AN", "2PN AN", "2PN avg AN", "1.5PN NUM", "2PN NUM", "2PN FIXED"};
    selectedFunctions = allFunctions⟦plotIndices⟧;
    TorbVal = Torb[data];
```

```
    tValues = Table[tmin + i * (tmax - tmin) / (numPoints - 1), {i, 0, numPoints - 1}];
    tScaledData = Table[Table[{t / TorbVal, selectedFunctions⟦i⟧[t, data]},
       {t, tValues}], {i, Length[selectedFunctions]}];
    ListLinePlot[tScaledData, Frame → True, FrameLabel →
      {"t / \!\(\*SubscriptBox[\(T\), \(orb\)]\)", LegendLtoJ⟦Compi, Compj⟧},
     PlotRange → All, FrameStyle → Directive[Black, Thin],
     TicksStyle → Directive[FontSize → 12],
     PlotStyle → {{Continue, CBlue}, {Continue, COrange},
        {Continue, CGreen}, {Dashed, CPurple}, {Dashed, CRed},
        {Continue, CBrown}, {Dashed, Brown}}⟦plotIndices⟧,
     PlotLegends → Placed[LineLegend[Style[#, FontSize → 16] & /@
         functionLabels⟦plotIndices⟧, LegendLayout → {"Row", 1}], Top]]];

DefPlotRLframe[data_, NumComp_, tmin_, tmax_, plotIndices_List, numPoints_ : 10] :=
  Module[{allFunctions, functionLabels,
     selectedFunctions, TorbVal, tValues, tScaledData},
    allFunctions =
     {PlotAn1, PlotAn2S, PlotAn2Savg, PlotNum1, PlotNum2S, PlotAn2SFixed};
functionLabels =
     {"1.5PN AN", "2PN AN", "2PN avg AN", "1.5PN NUM", "2PN NUM", "2PN FIXED"};
    selectedFunctions = allFunctions⟦plotIndices⟧;
    TorbVal = Torb[data];
    tValues = Table[tmin + i * (tmax - tmin) / (numPoints - 1), {i, 0, numPoints - 1}];
    tScaledData = Table[Table[{t / TorbVal, selectedFunctions⟦i⟧[t, data]},
       {t, tValues}], {i, Length[selectedFunctions]}];
    ListLinePlot[tScaledData, Frame → True,
     FrameLabel → {"t / \!\(\*SubscriptBox[\(T\), \(orb\)]\)",
       {"Rx in Lframe", "Ry in Lframe", "Rz in Lframe"}⟦NumComp⟧},
     PlotRange → All, FrameStyle → Directive[Black, Thin],
     TicksStyle → Directive[FontSize → 12],
     PlotStyle → {{Continue, CBlue}, {Continue, COrange}, {Continue, CGreen},
        {Dashed, CPurple}, {Dashed, CRed}, {Dashed, CBrown}}⟦plotIndices⟧,
     PlotLegends → Placed[LineLegend[Style[#, FontSize → 16] & /@
         functionLabels⟦plotIndices⟧, LegendLayout → {"Row", 1}], Top]]];
```

In[ ]:=

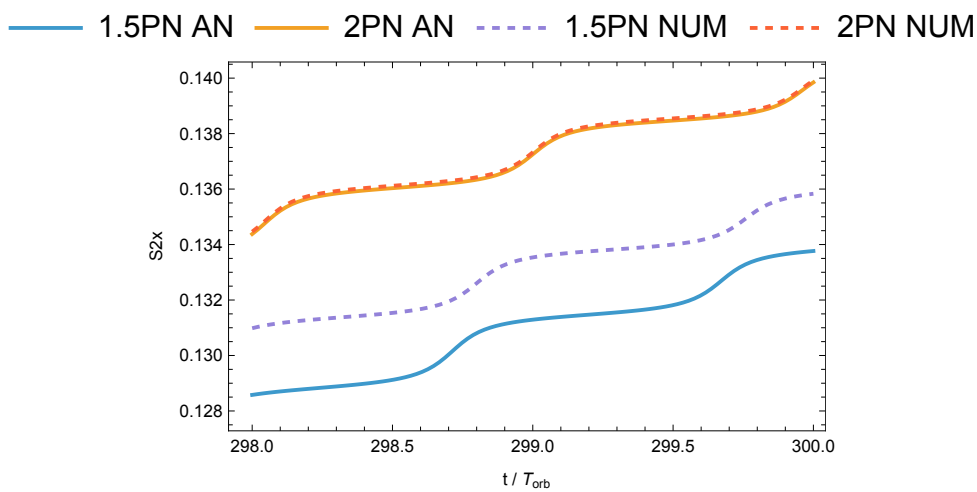
In[ ]:=

B) Plots

In[ ]:=
```
(*TEST S2 SOLUTION : OK*)
data = data1; NumVec = 4; NumComp = 1;
tmin = 298 Torb[data];
tmax = tmin +  2 Torb[data];
plotIndices = {1, 2, 4, 5(*,6*)};
Npoints = 100;

DefPlot[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

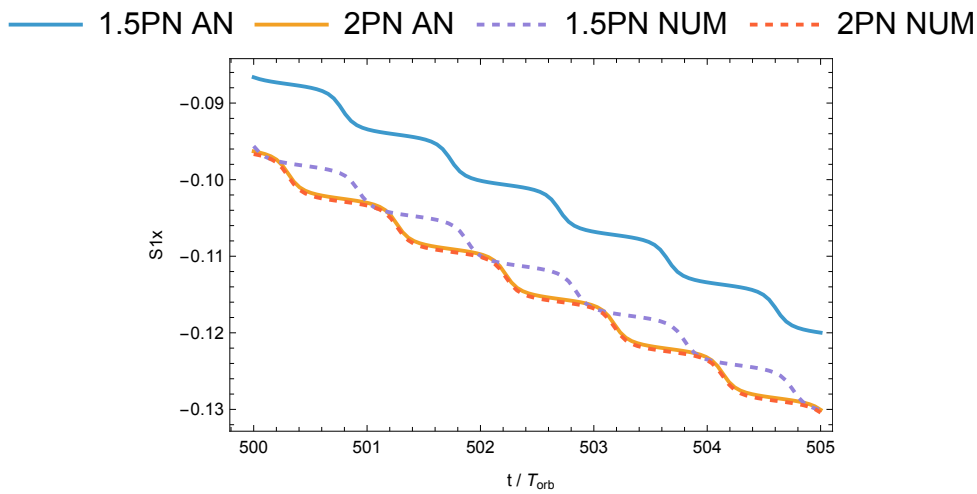
Out[ ]=



In[ ]:=
```
(*TEST S1 SOLUTION : OK*)
data = data1; NumVec = 3; NumComp = 1;
tmin = 500 Torb[data];
tmax = tmin + 5 Torb[data];
plotIndices = {1, 2, 4, 5(*,6*)};
Npoints = 100;

DefPlot[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

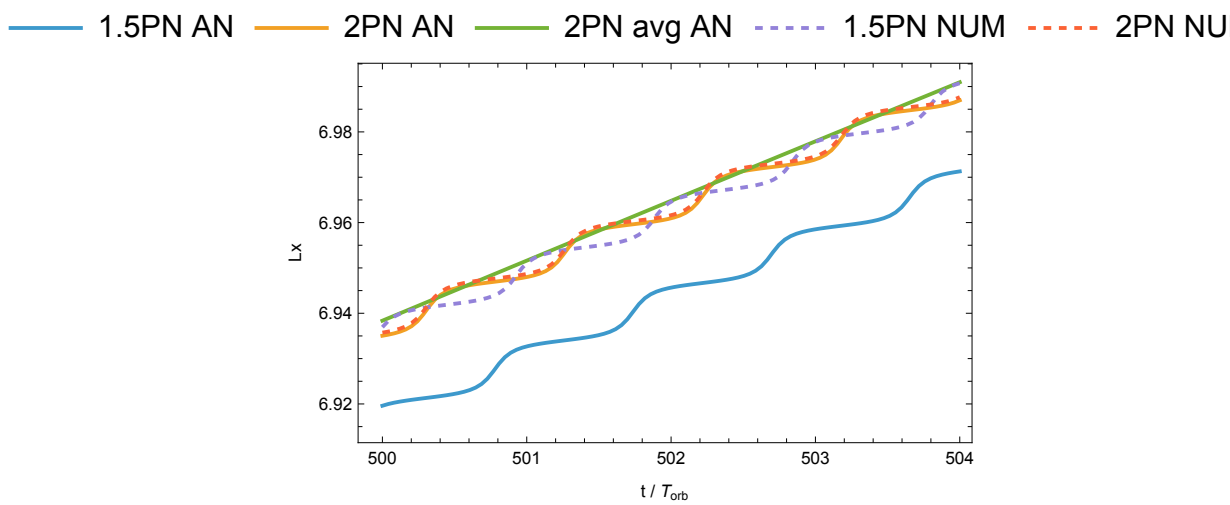
Out[ ]=



In[ ]:=
```
(*TEST L SOLUTION : OK*)
data = data1; NumVec = 2; NumComp = 1;
tmin = 500 Torb[data];
tmax = tmin + 4 Torb[data];
plotIndices = {1, 2, 3, 4, 5(*,6*)};
Npoints = 100;

DefPlot[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

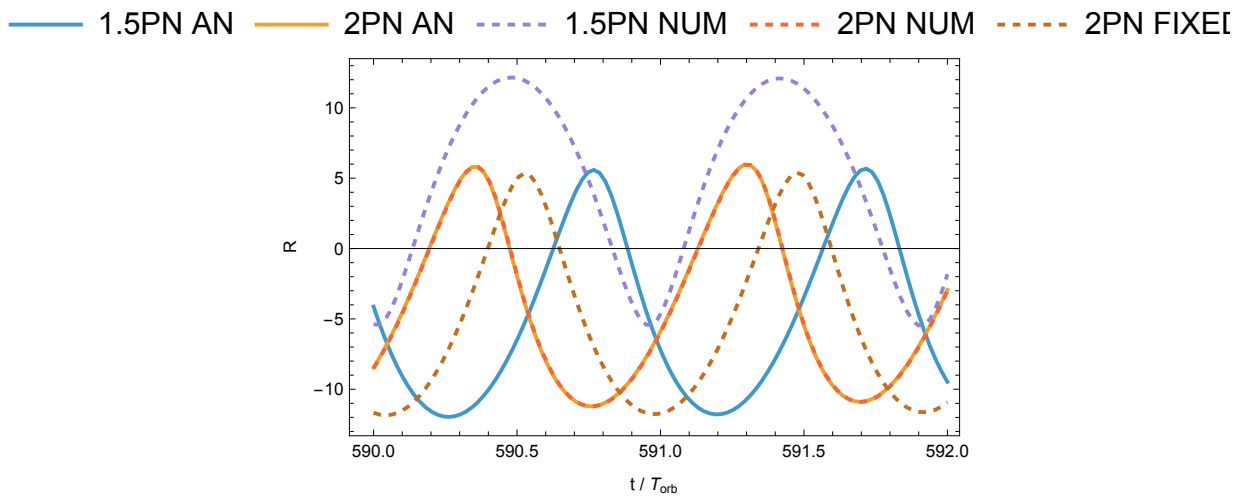
Out[ ]=



In[ ]:=
```
(*TEST Rmag SOLUTION : OK*)
data = data1; NumVec = 1; NumComp = 3;
tmin =  590 Torb[data];
tmax = tmin + 2 Torb[data];
plotIndices = {1, 2, 4, 5, 6};
Npoints = 100;

DefPlotNorm[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

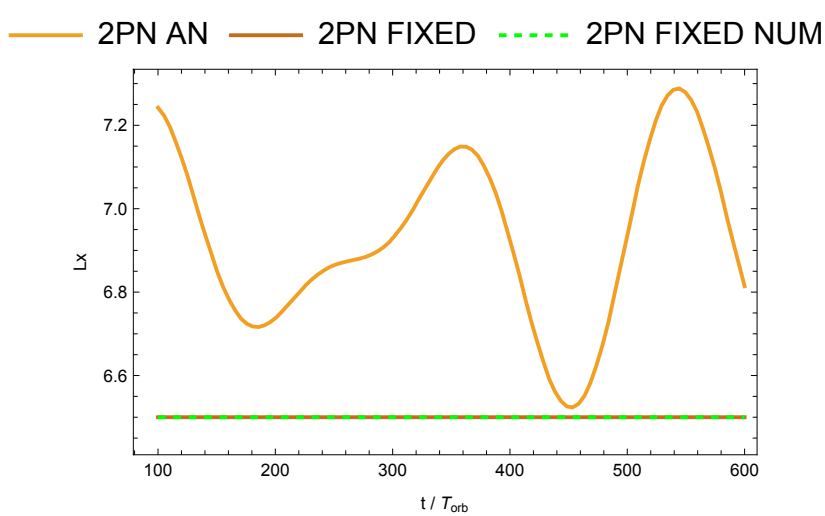
Out[ ]=



In[ ]:=
```
data = data1; NumVec = 2; NumComp = 1;
tmin = 100 Torb[data];
tmax = tmin + 500 Torb[data];
plotIndices = {2, 6, 7};
Npoints = 100;

DefPlot[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

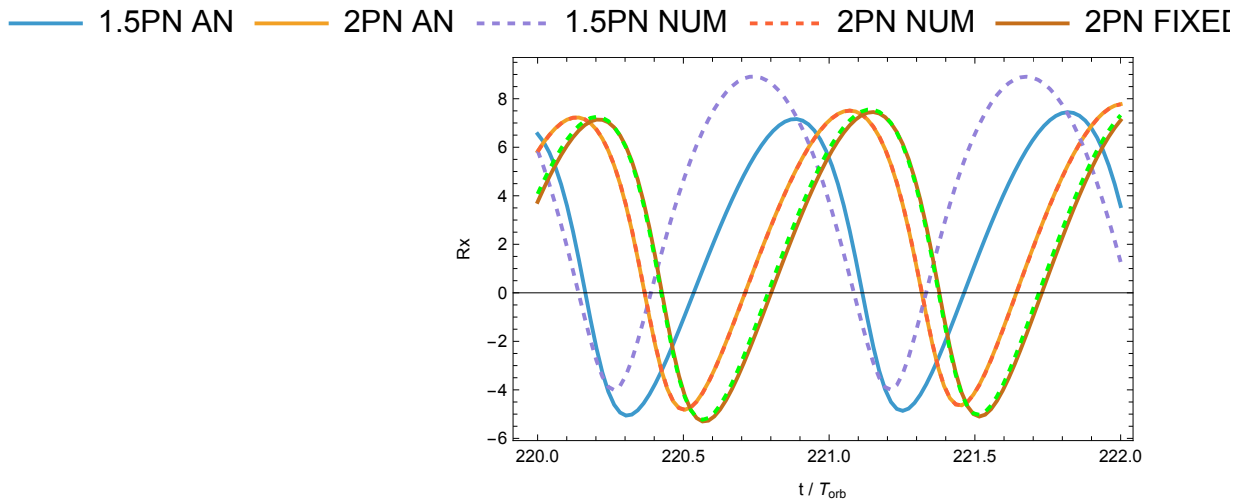
Out[ ]=



In[ ]:=
```
data = data1; NumVec = 1; NumComp = 1;
tmin = 220 Torb[data];
tmax = tmin + 2 Torb[data];
plotIndices = {1, 2, 4, 5, 6, 7};
Npoints = 100;

DefPlot[data, NumVec, NumComp, tmin, tmax, plotIndices, Npoints]
```

Out[ ]=



In[ ]:=

In[ ]:=

In[ ]:=
```
Speak["Job done"]
```